

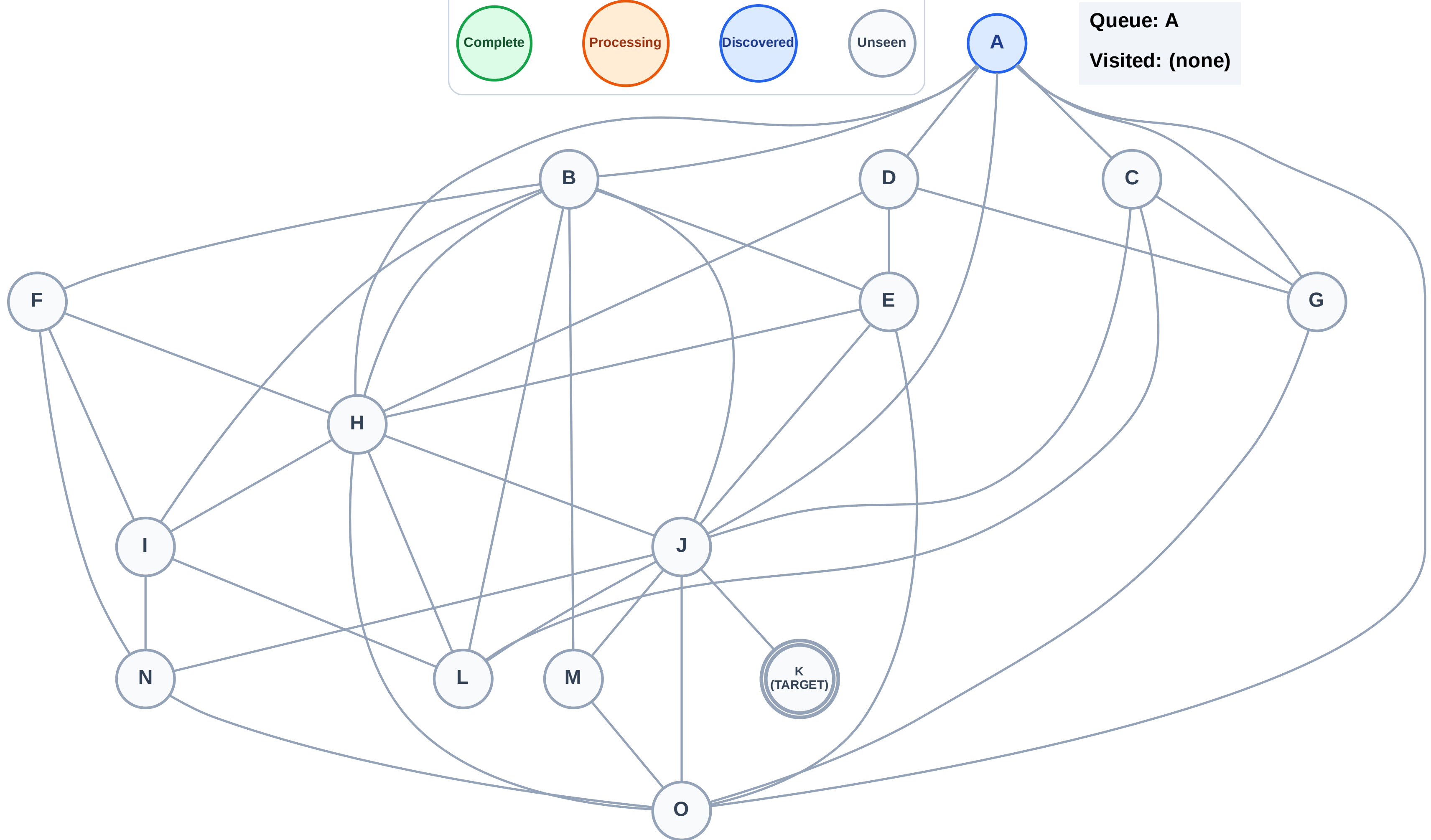
# BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

## Legend



Queue: A  
Visited: (none)



# BFS Traversal

Step 2: Process node A

Legend

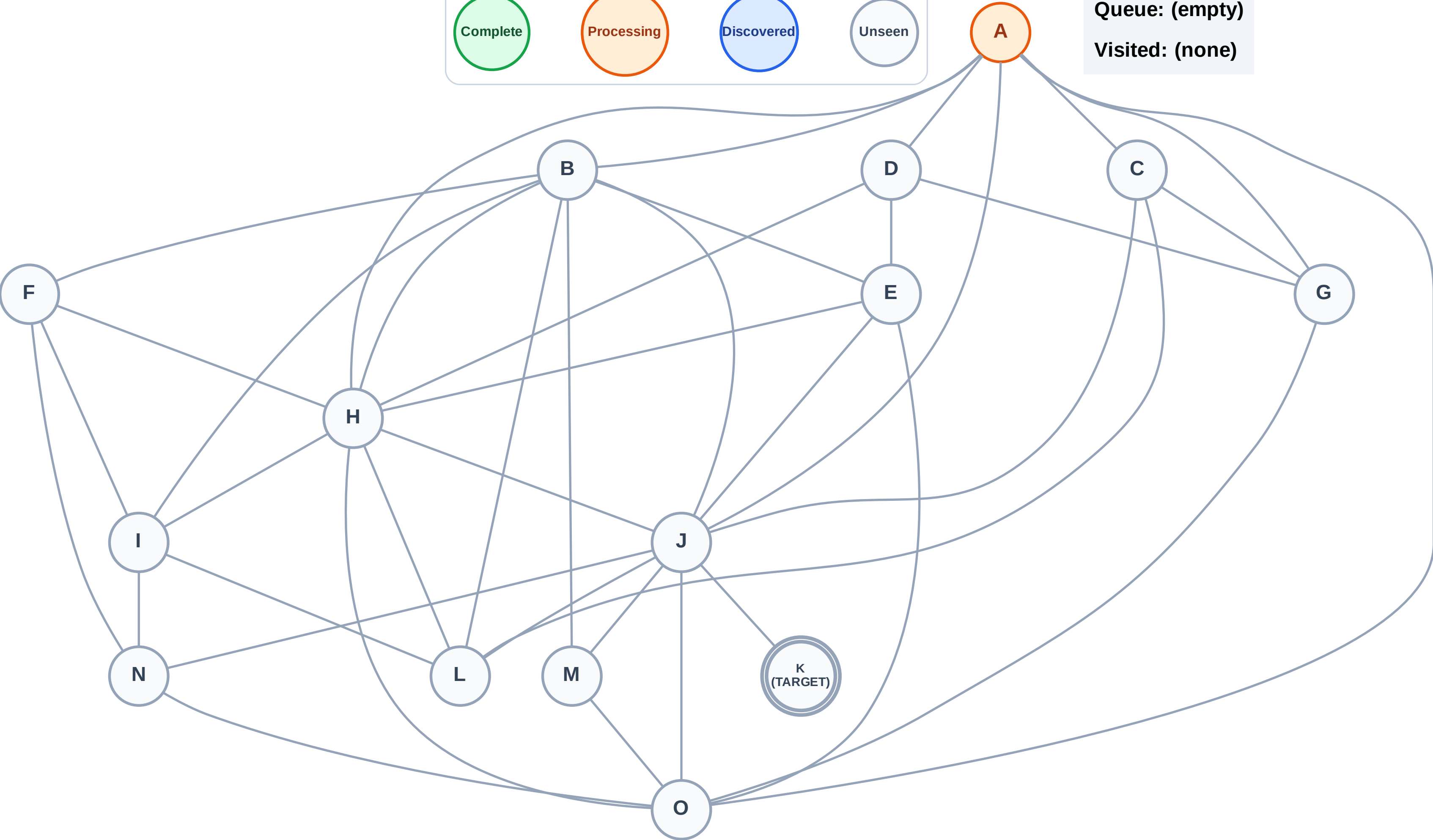
Complete

Processing

Discovered

Unseen

Queue: (empty)  
Visited: (none)



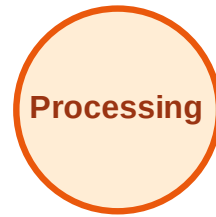
# BFS Traversal

Step 3: Discover B, C, D, G, H, J, O from A

## Legend



Complete



Processing



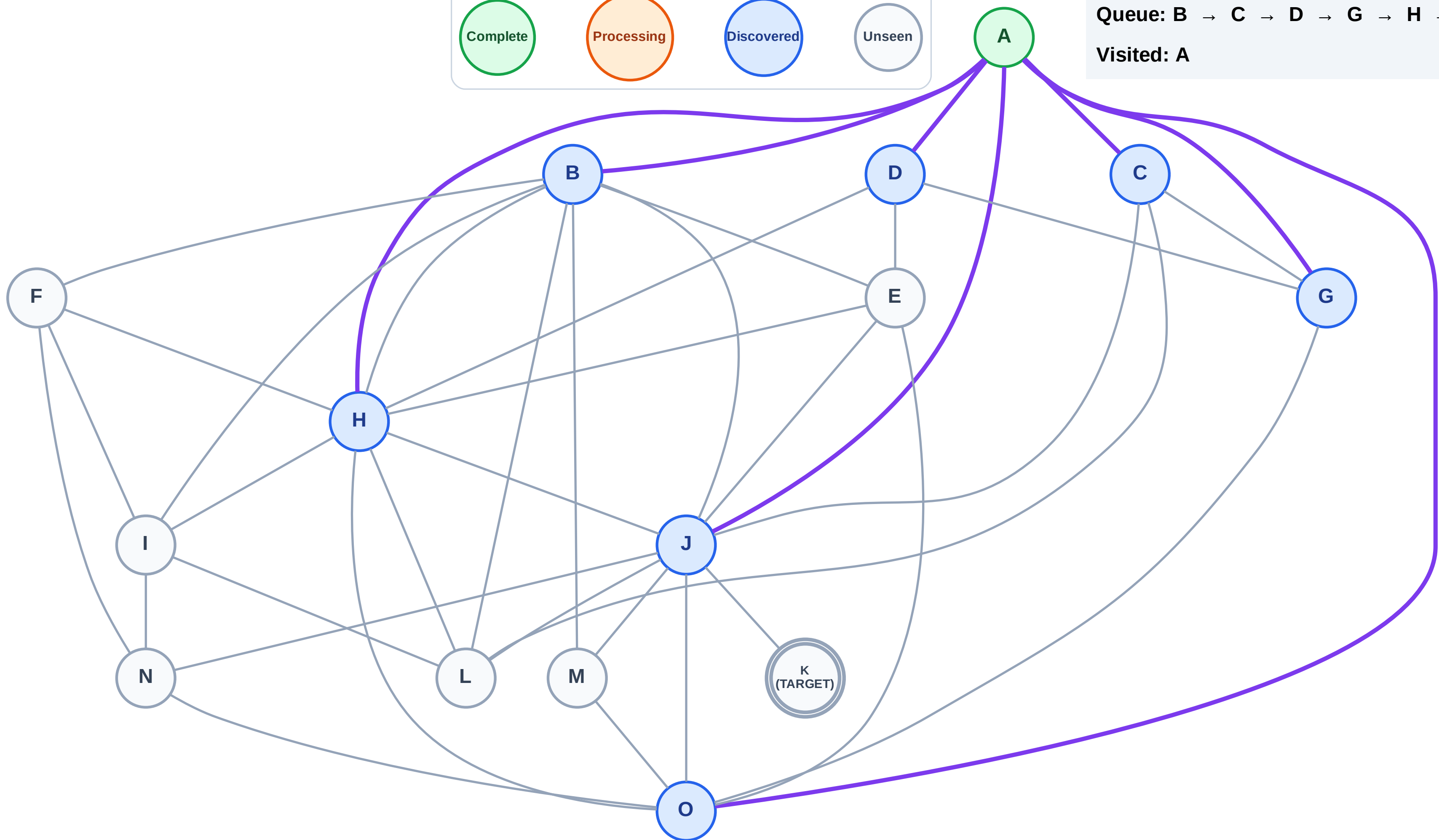
Discovered



Unseen

Queue: B → C → D → G → H → J → O

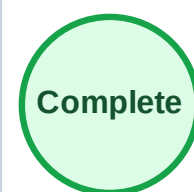
Visited: A



# BFS Traversal

Step 4: Process node B

## Legend



Complete



Processing

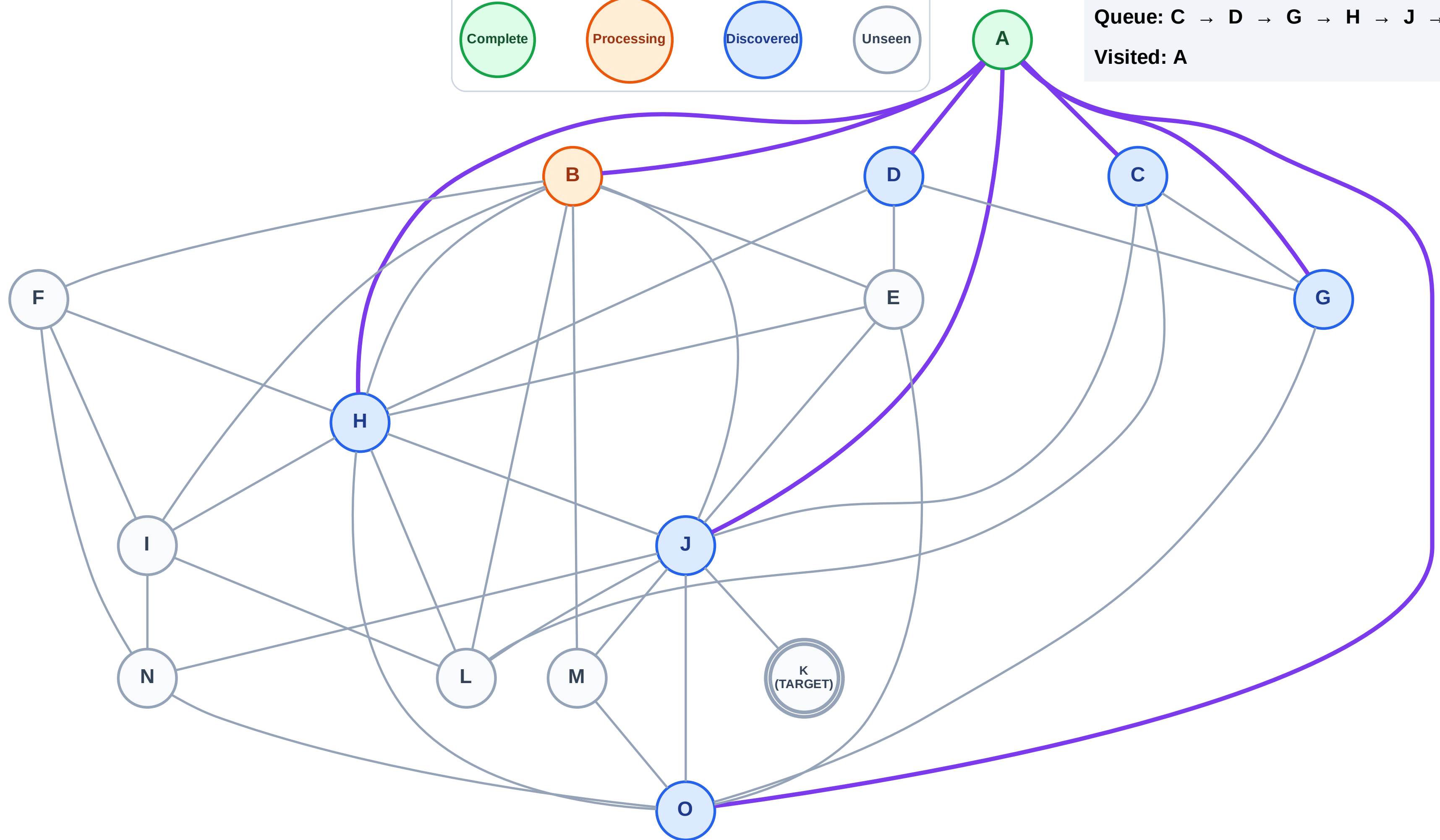


Discovered



Unseen

Queue: C → D → G → H → J → O  
Visited: A



# BFS Traversal

Step 5: Discover E, F, I, L, M from B

## Legend



Complete



Processing



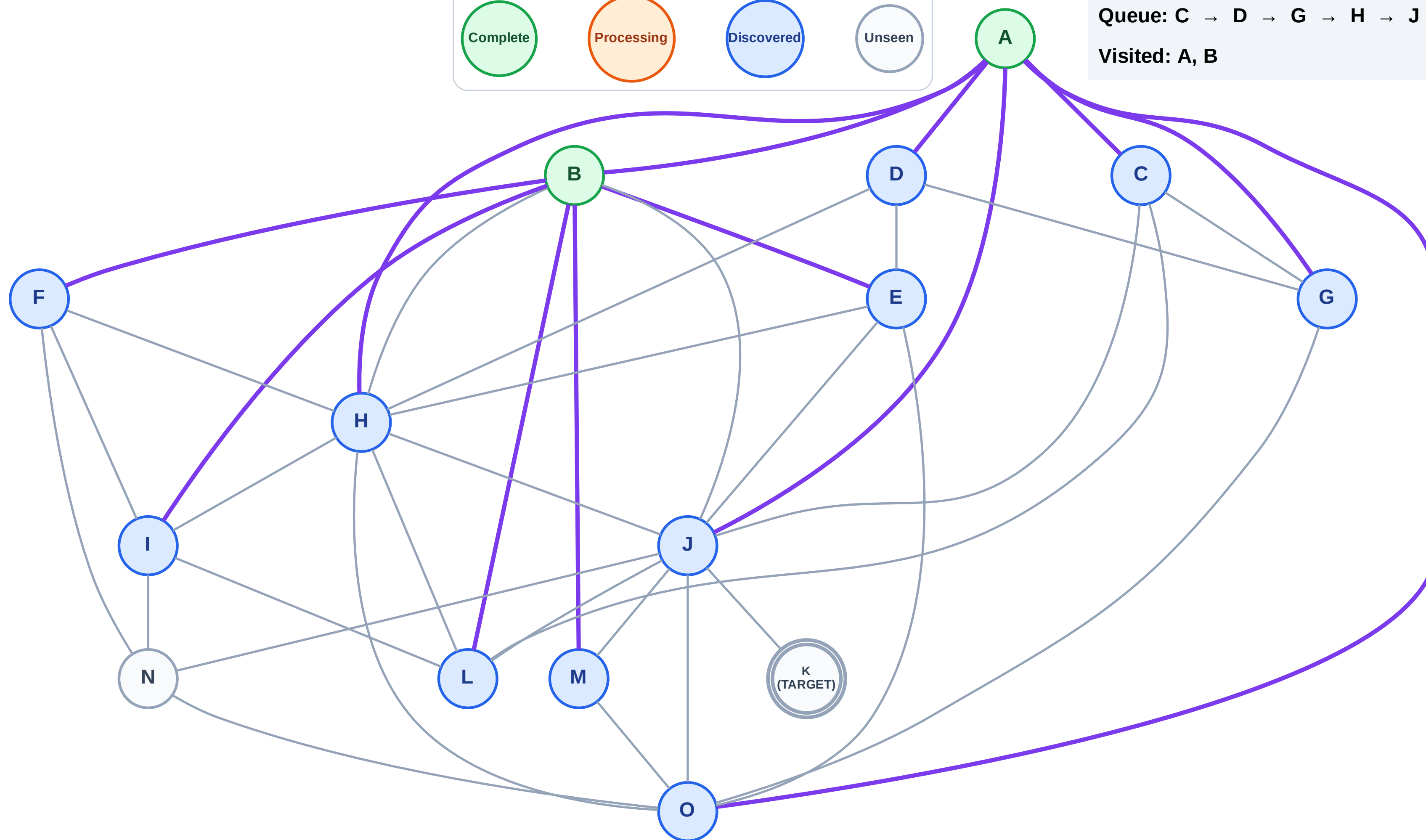
Discovered



Unseen

Queue: C → D → G → H → J → O → E → F → I → L → M

Visited: A, B



# BFS Traversal

Step 6: Process node C

## Legend



Complete



Processing



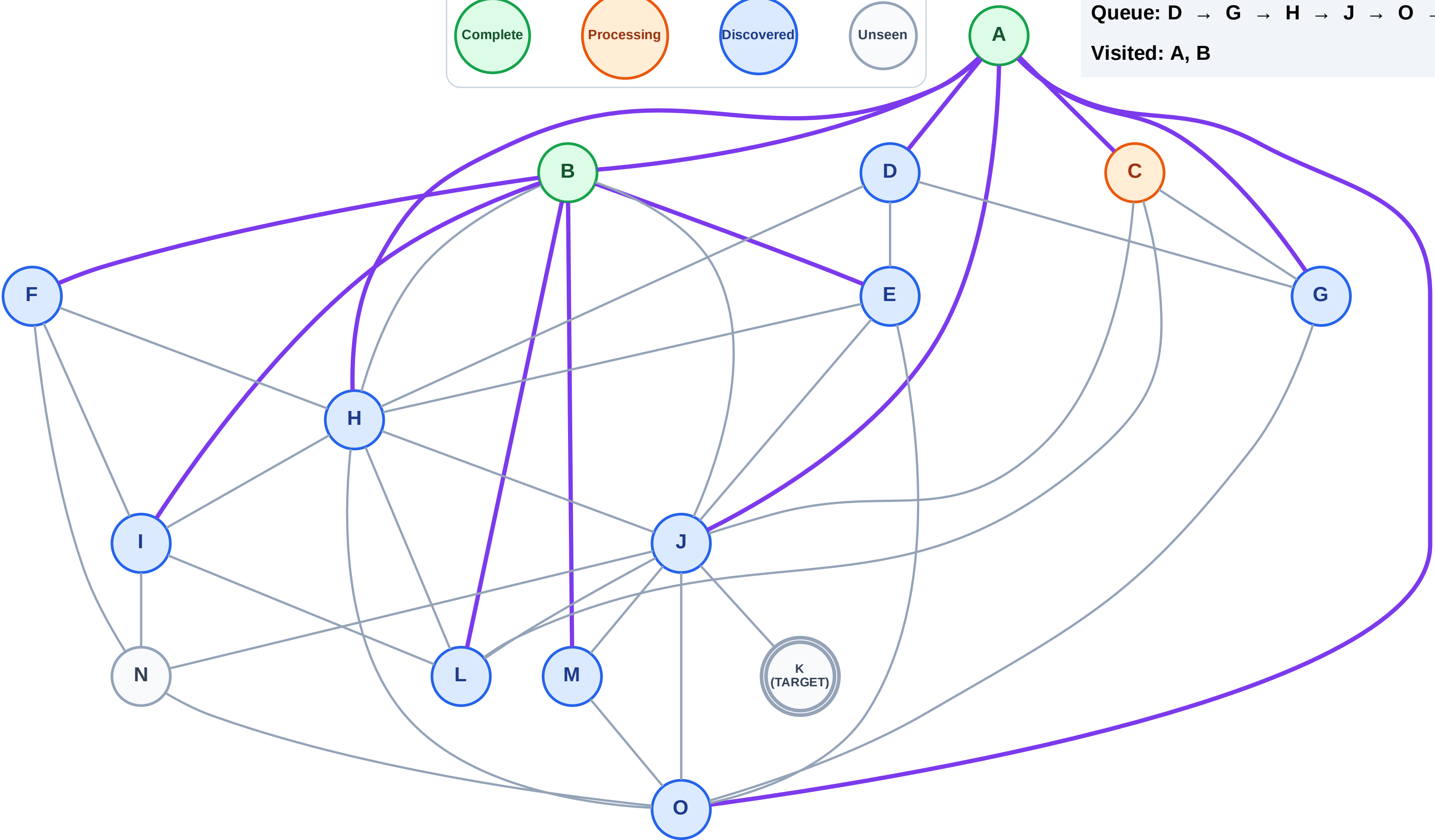
Discovered



Unseen

Queue: D → G → H → J → O → E → F → I → L → M

Visited: A, B



# BFS Traversal

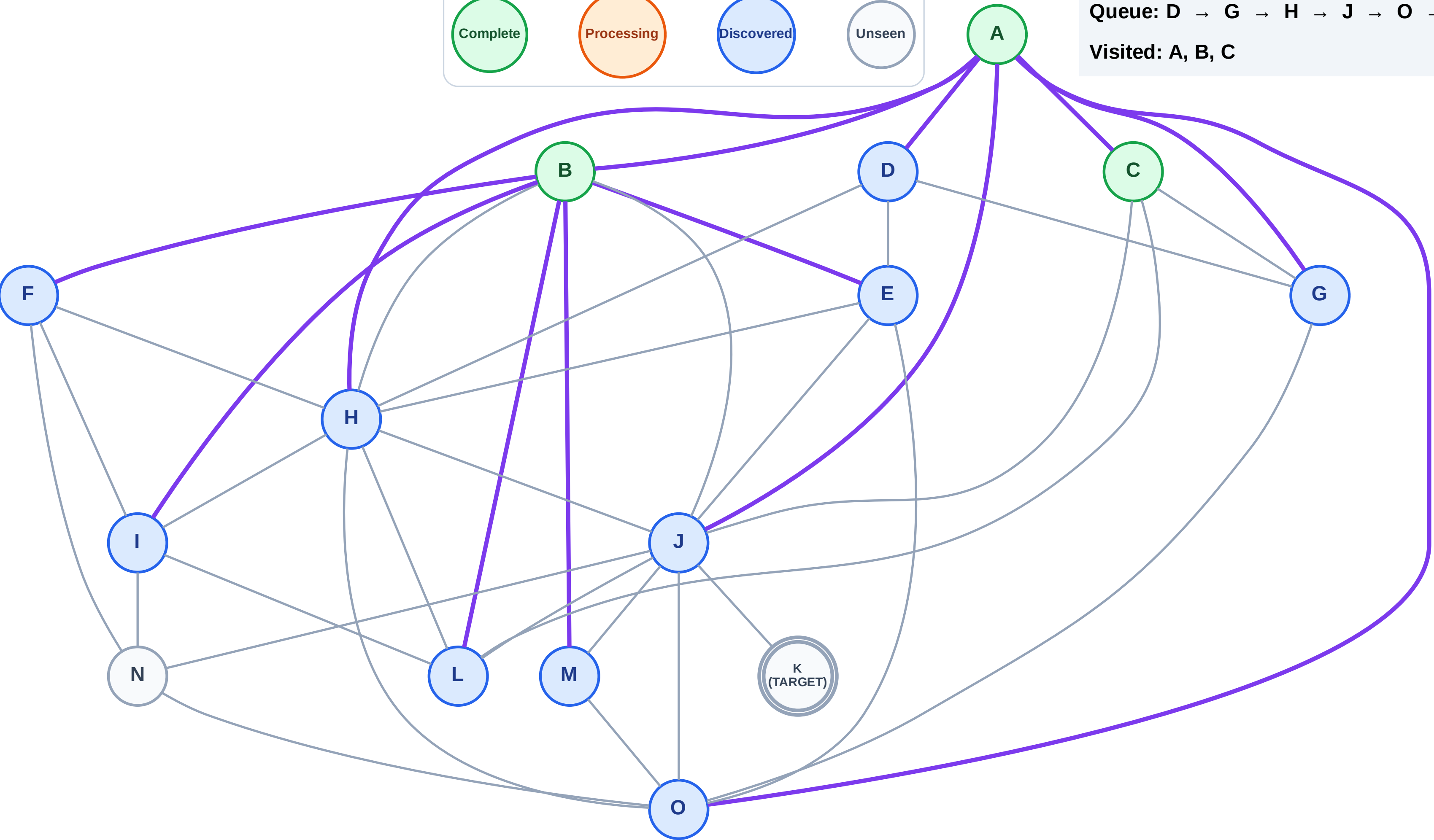
Step 7: No new neighbors from C

## Legend



Queue: D → G → H → J → O → E → F → I → L → M

Visited: A, B, C



# BFS Traversal

Step 8: Process node D

Legend

Complete

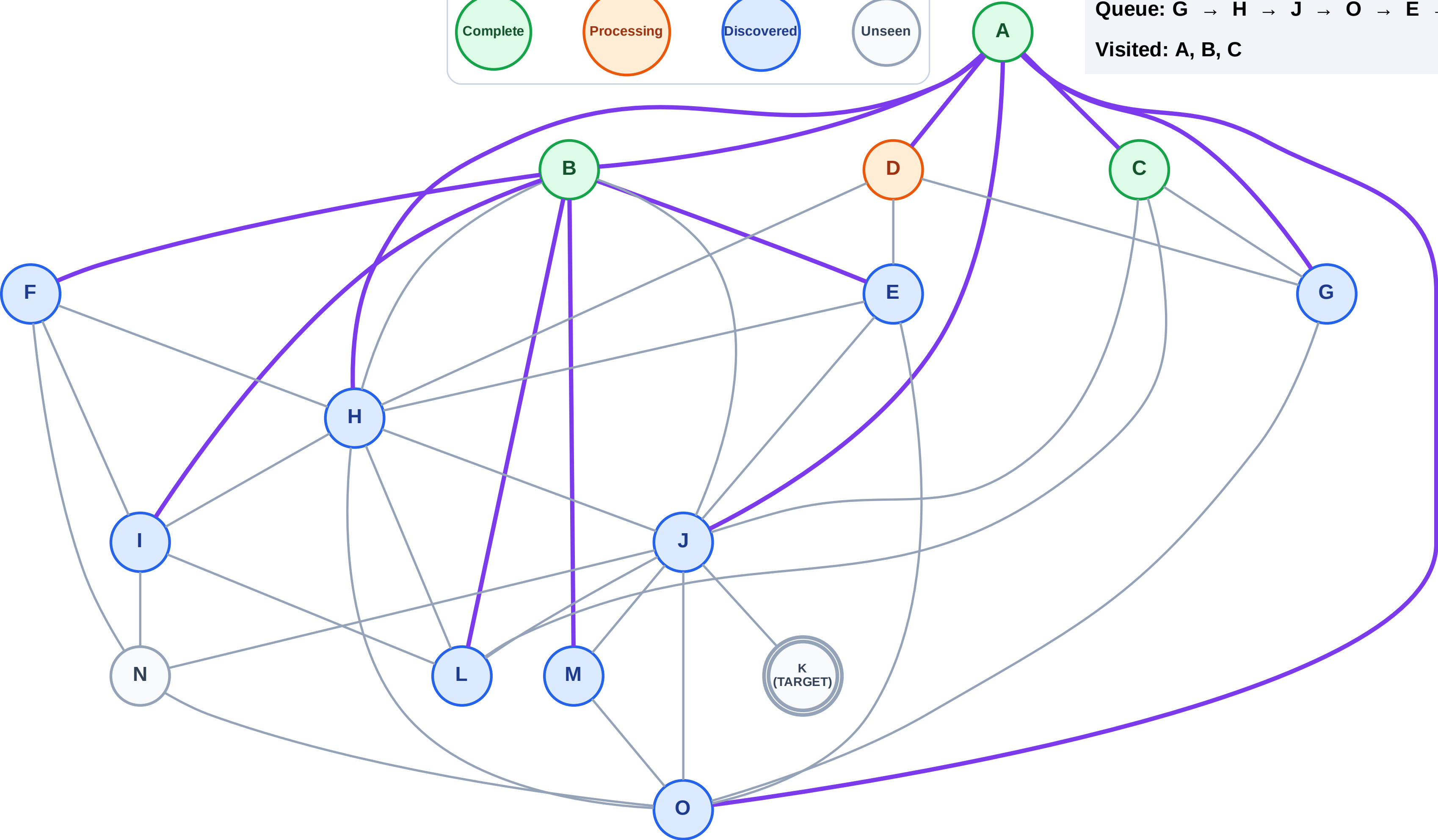
Processing

Discovered

Unseen

Queue: G → H → J → O → E → F → I → L → M

Visited: A, B, C



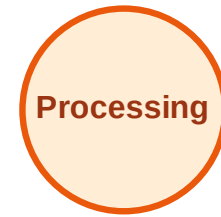
# BFS Traversal

Step 9: No new neighbors from D

## Legend



Complete



Processing



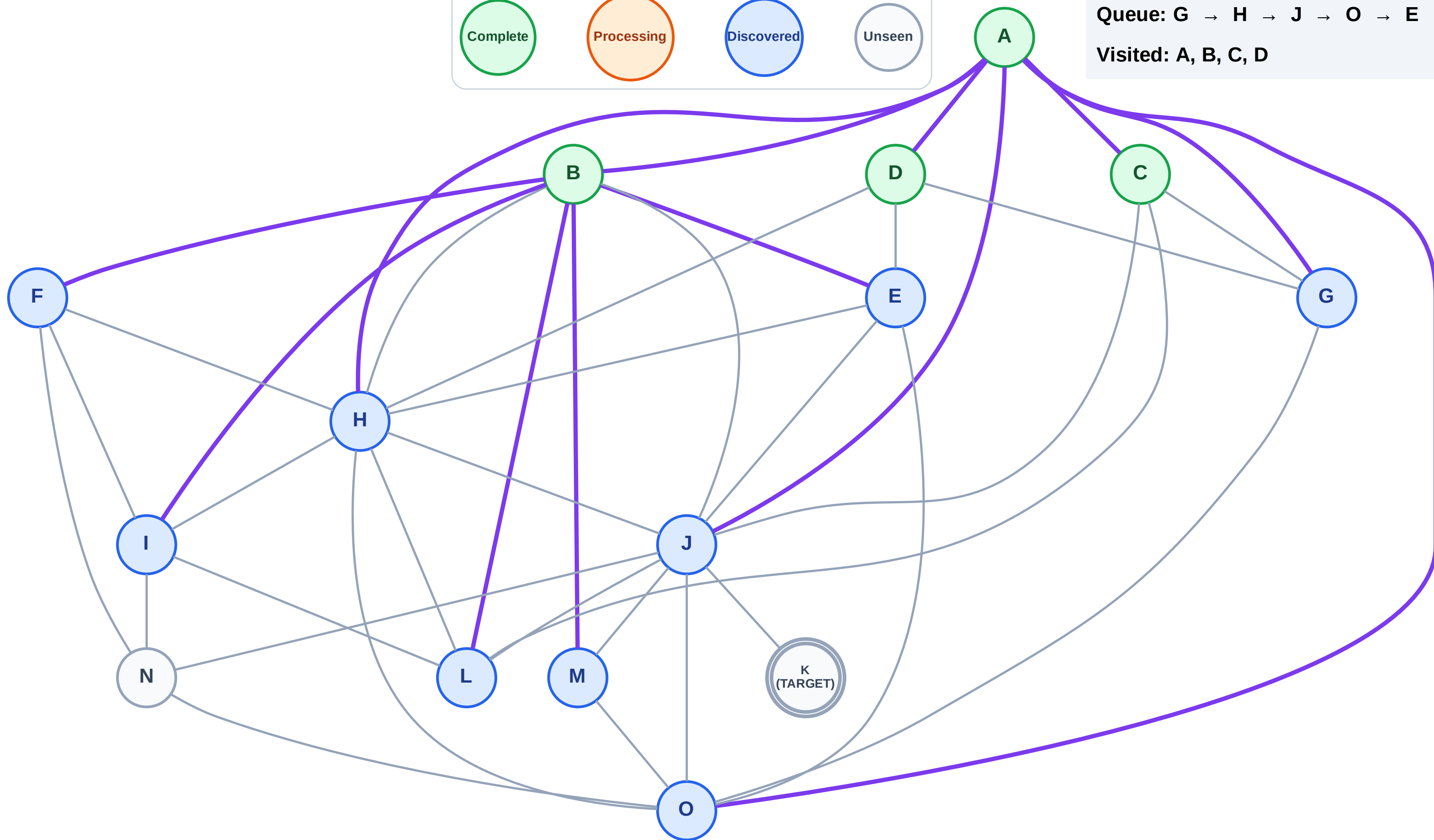
Discovered



Unseen

Queue: G → H → J → O → E → F → I → L → M

Visited: A, B, C, D



# BFS Traversal

Step 10: Process node G

## Legend



Complete



Processing



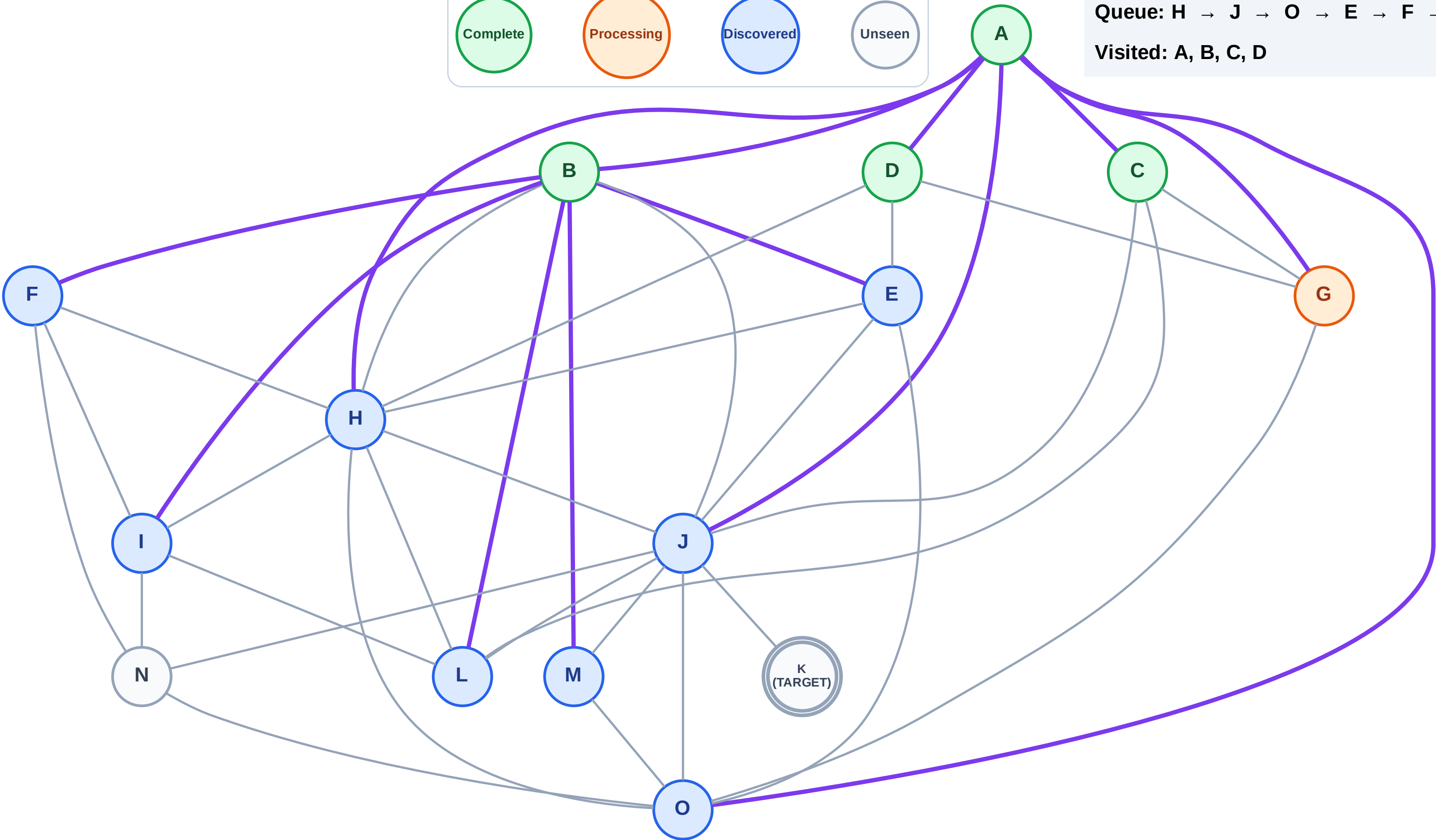
Discovered



Unseen

Queue: H → J → O → E → F → I → L → M

Visited: A, B, C, D



# BFS Traversal

Step 11: No new neighbors from G

## Legend



Complete



Processing



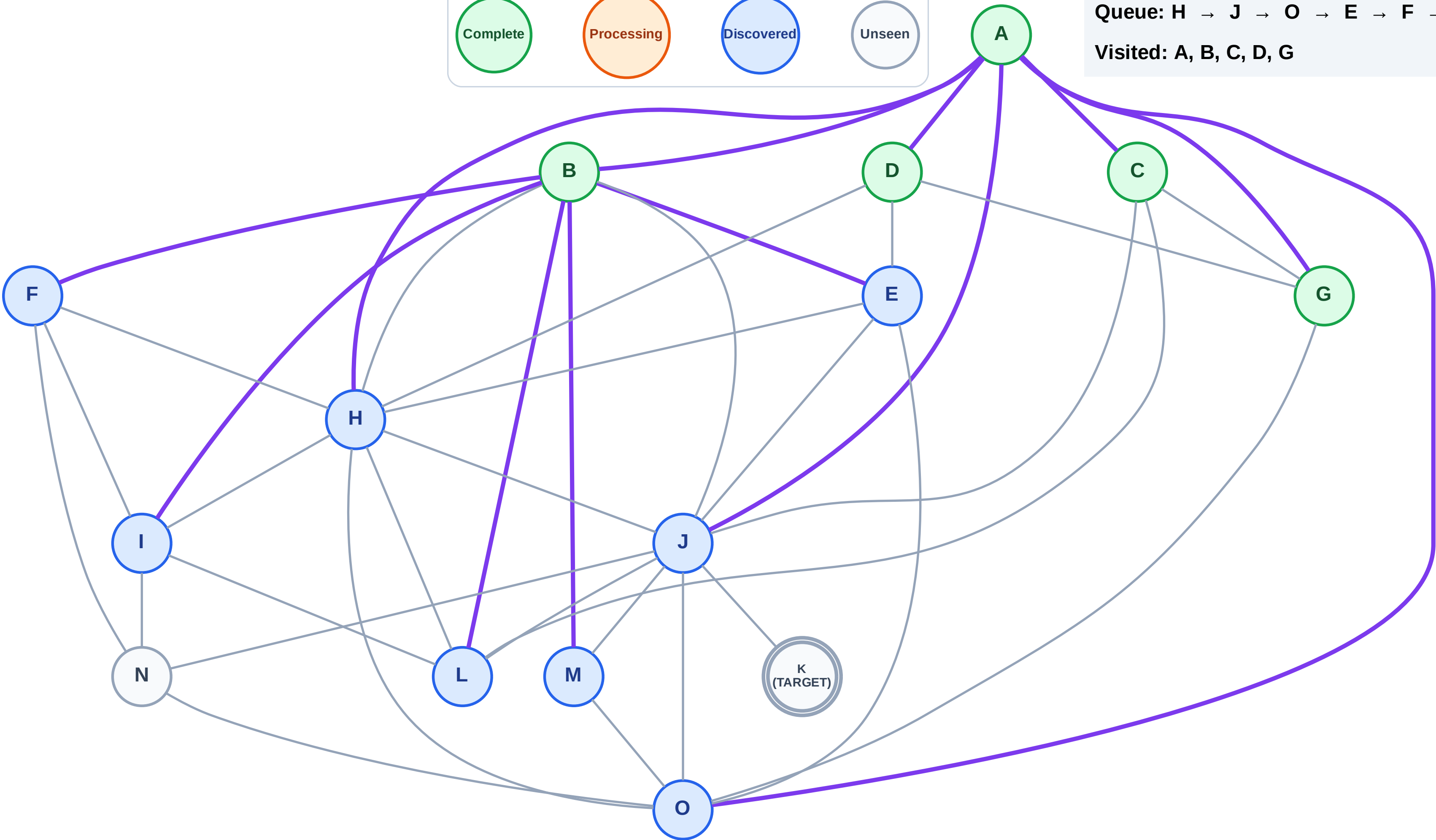
Discovered



Unseen

Queue: H → J → O → E → F → I → L → M

Visited: A, B, C, D, G



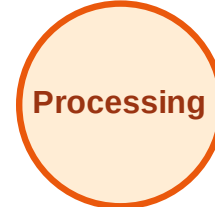
# BFS Traversal

Step 12: Process node H

## Legend



Complete



Processing



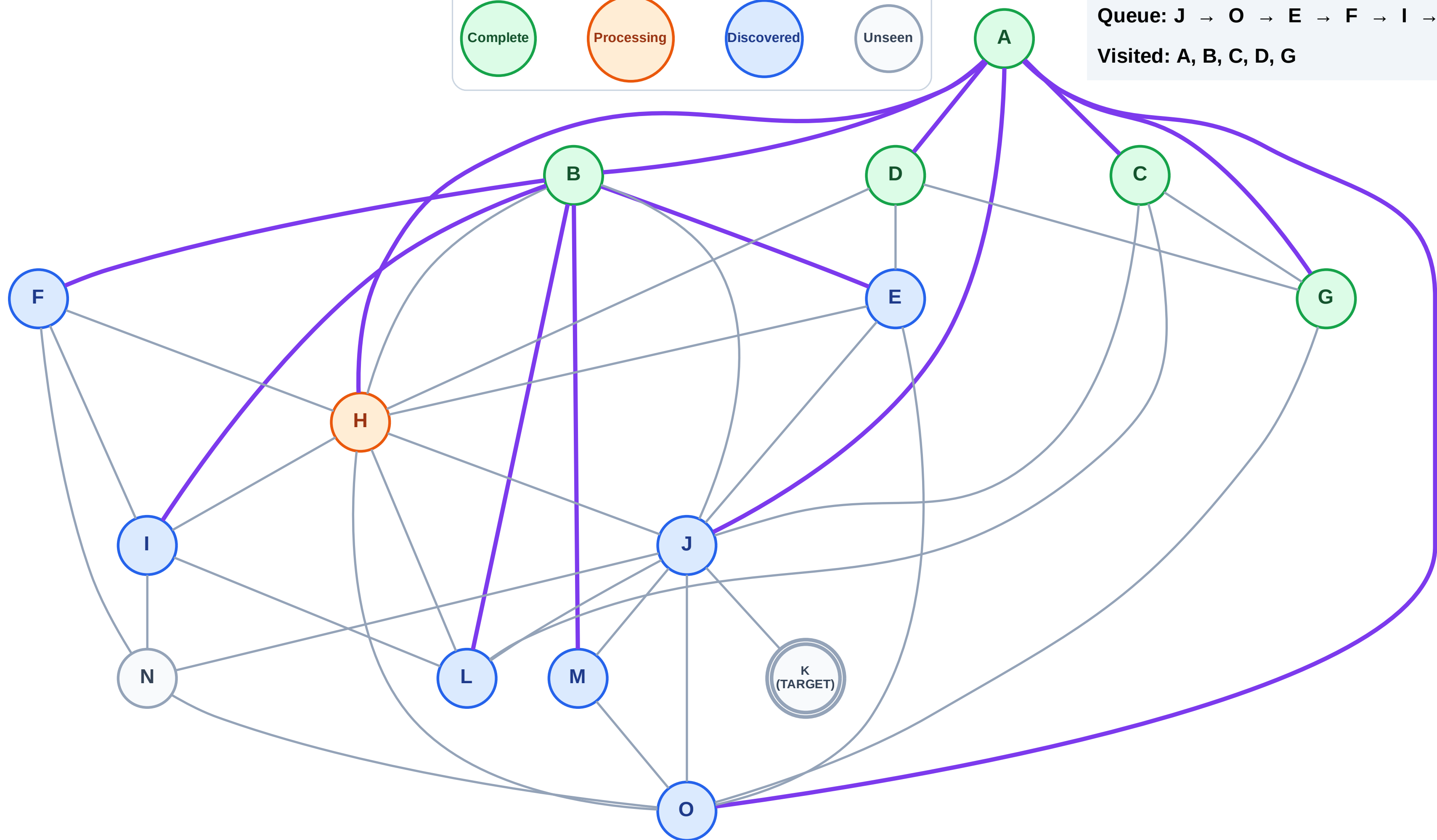
Discovered



Unseen

Queue: J → O → E → F → I → L → M

Visited: A, B, C, D, G



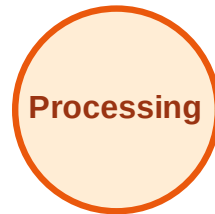
# BFS Traversal

Step 13: No new neighbors from H

## Legend



Complete



Processing



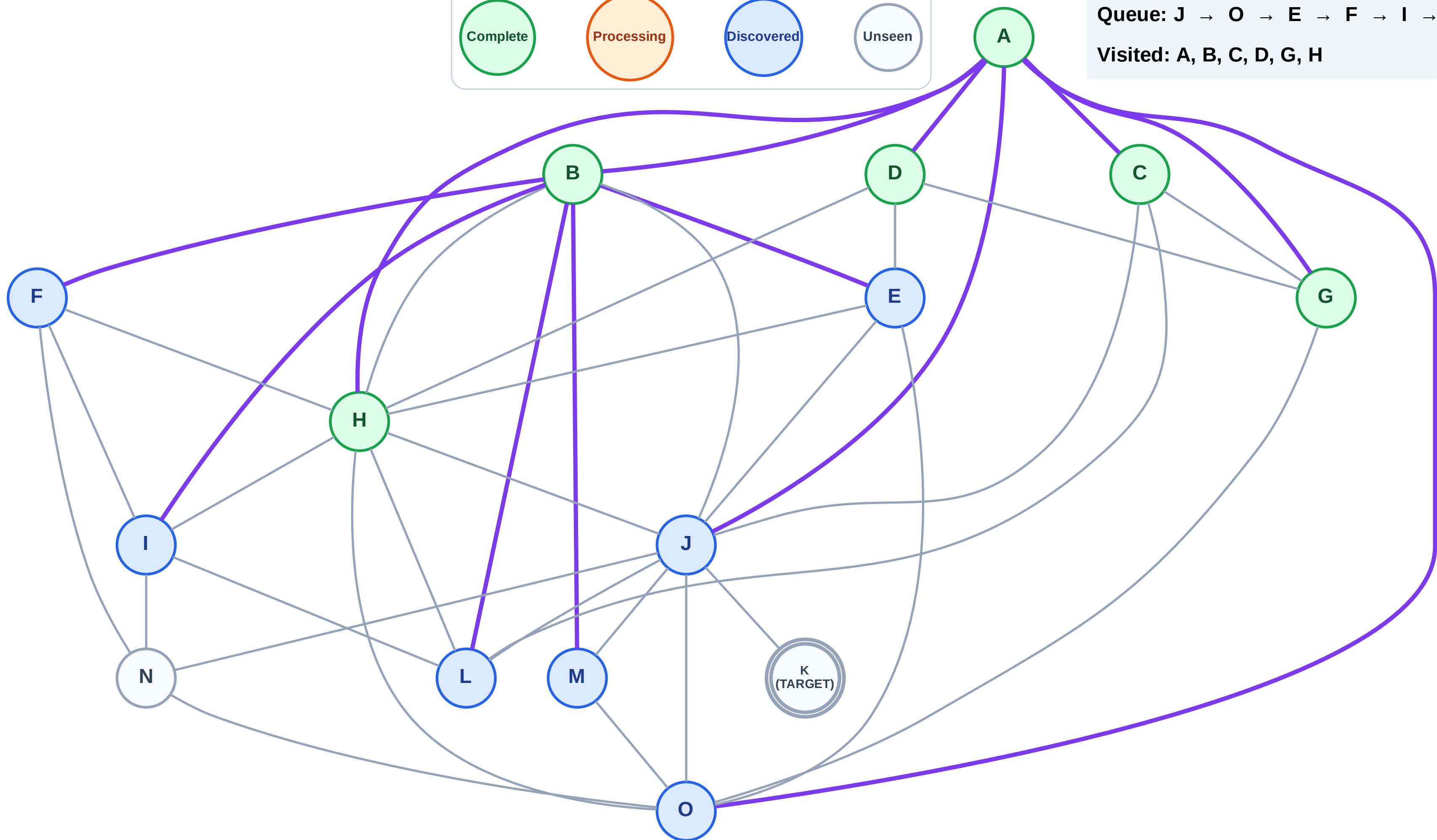
Discovered



Unseen

Queue: J → O → E → F → I → L → M

Visited: A, B, C, D, G, H



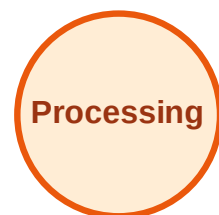
# BFS Traversal

Step 14: Process node J

## Legend



Complete



Processing



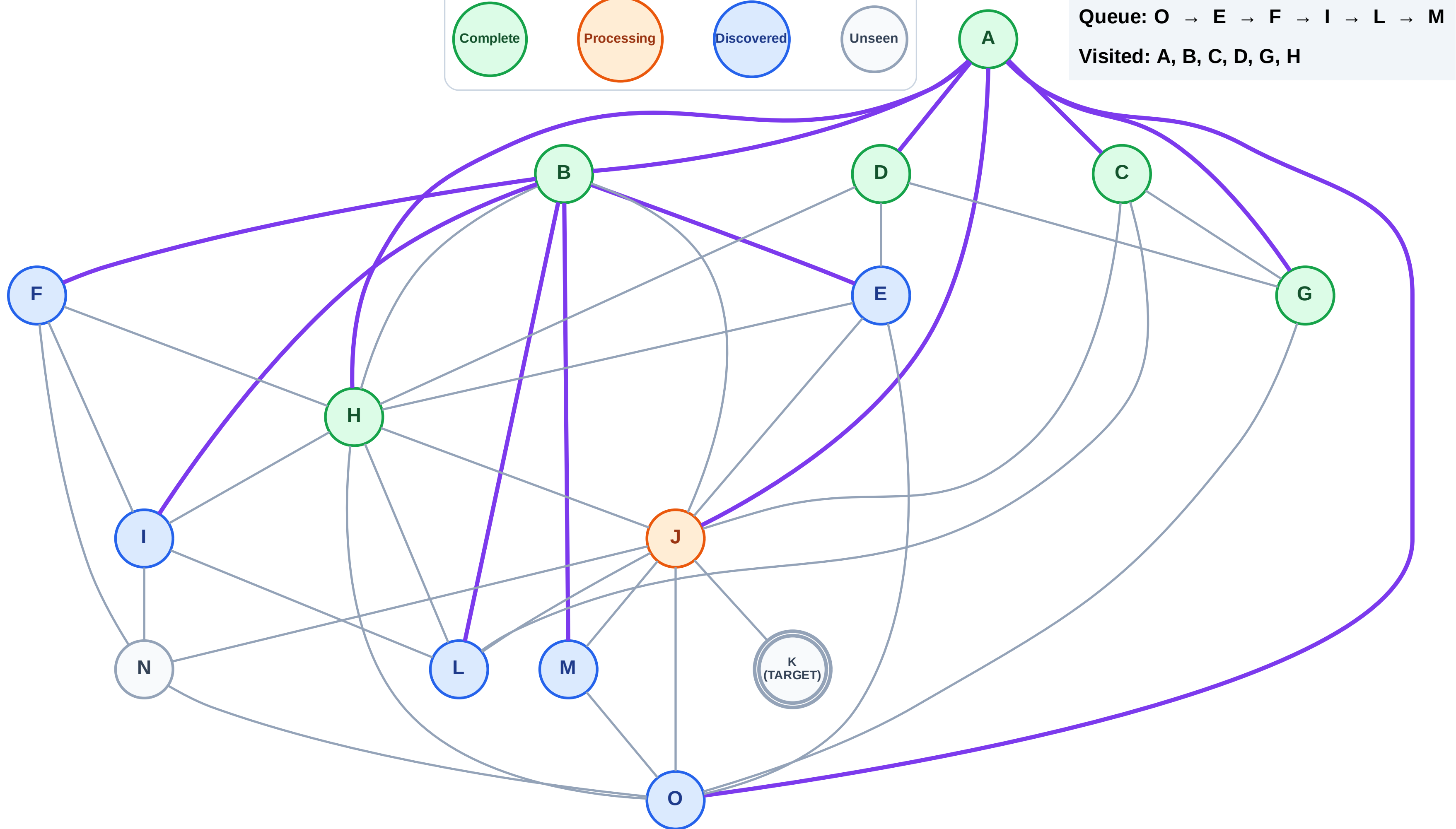
Discovered



Unseen

Queue: O → E → F → I → L → M

Visited: A, B, C, D, G, H



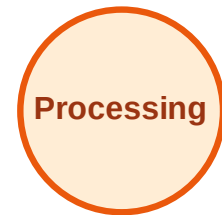
# BFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → J → K

## Legend



Complete



Processing



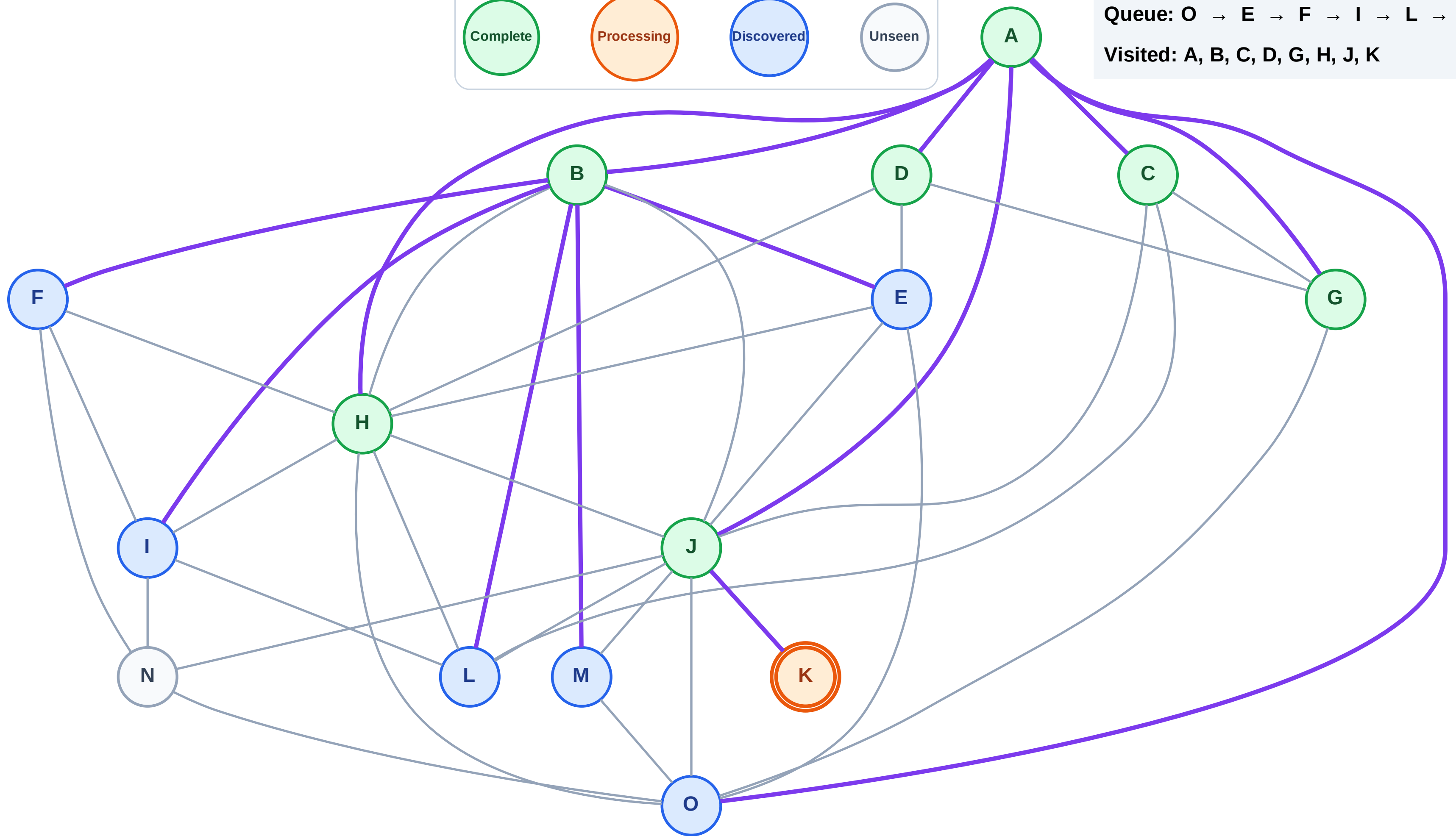
Discovered



Unseen

Queue: O → E → F → I → L → M

Visited: A, B, C, D, G, H, J, K



# DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

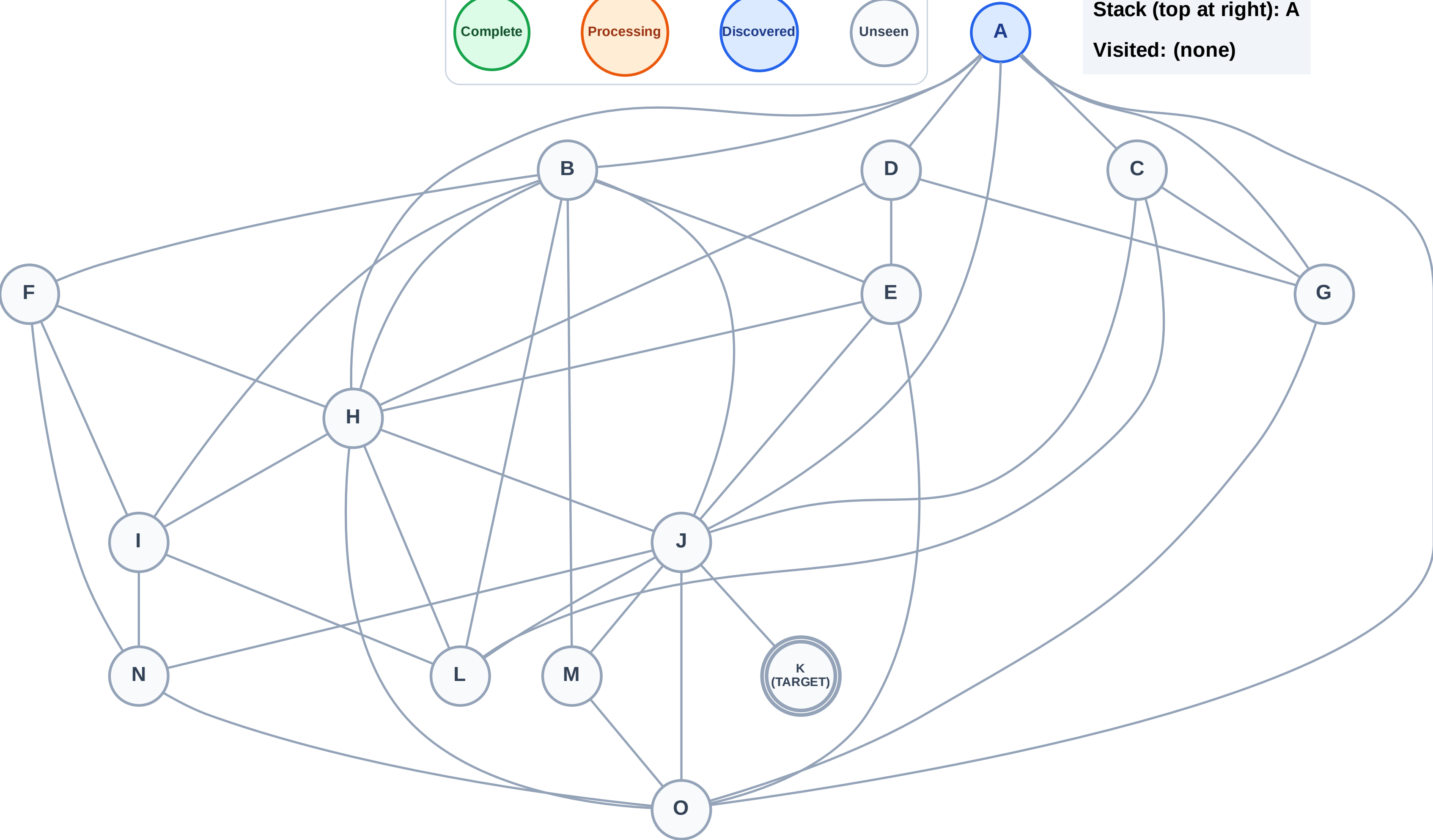
Complete

Processing

Discovered

Unseen

Stack (top at right): A  
Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

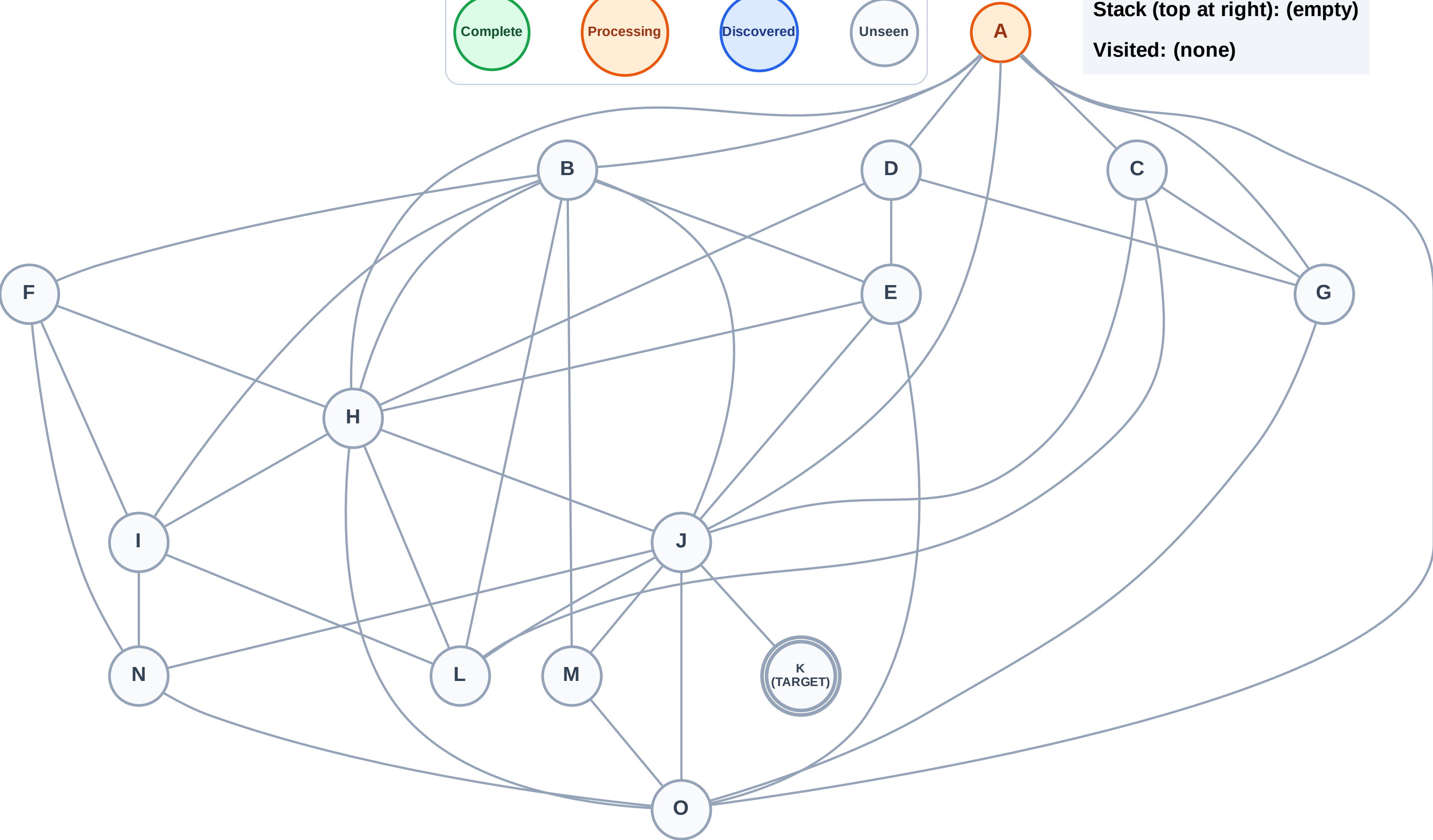
Complete

Processing

Discovered

Unseen

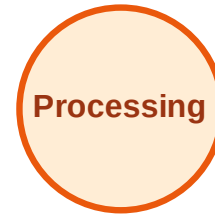
Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

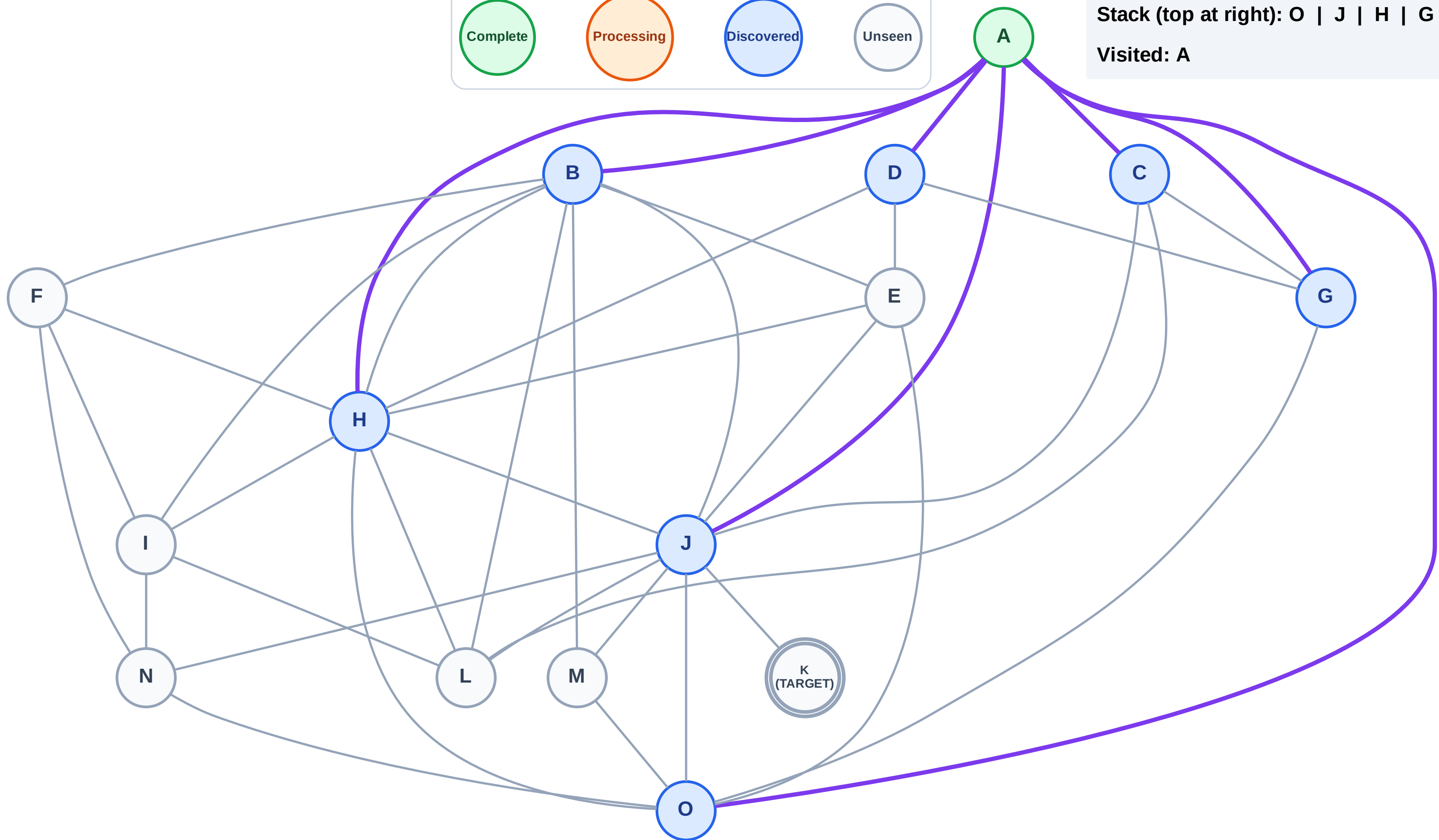
Step 3: Discover O, J, H, G, D, C, B from A

## Legend



Stack (top at right): O | J | H | G | D | C | B

Visited: A



# DFS Traversal

Step 4: Process node B

Legend

Complete

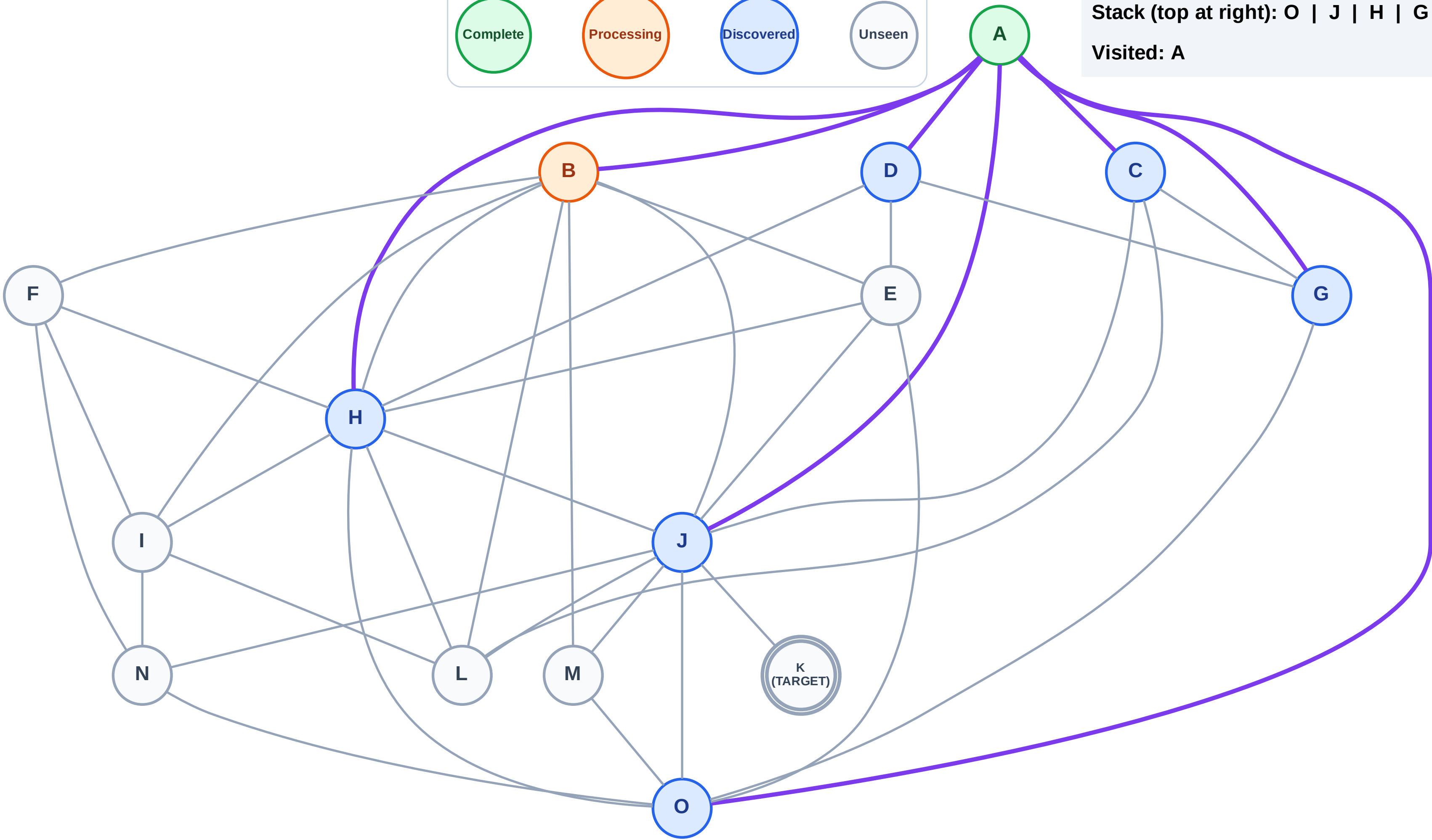
Processing

Discovered

Unseen

Stack (top at right): O | J | H | G | D | C

Visited: A



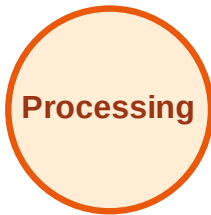
# DFS Traversal

Step 5: Discover M, L, I, F, E from B

## Legend



Complete



Processing



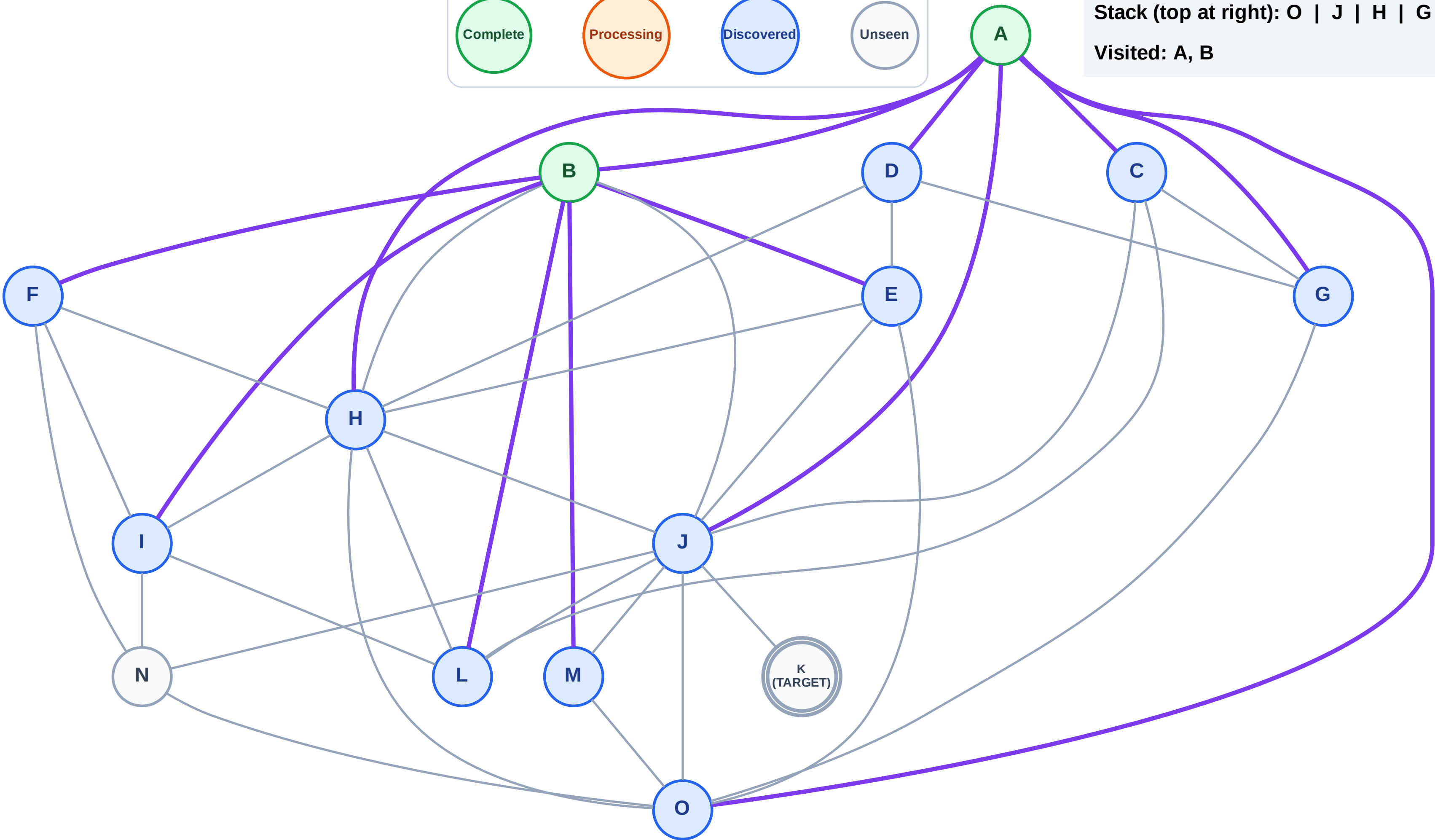
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C | M | L | I | F | E

Visited: A, B



# DFS Traversal

Step 6: Process node E

## Legend



Complete



Processing



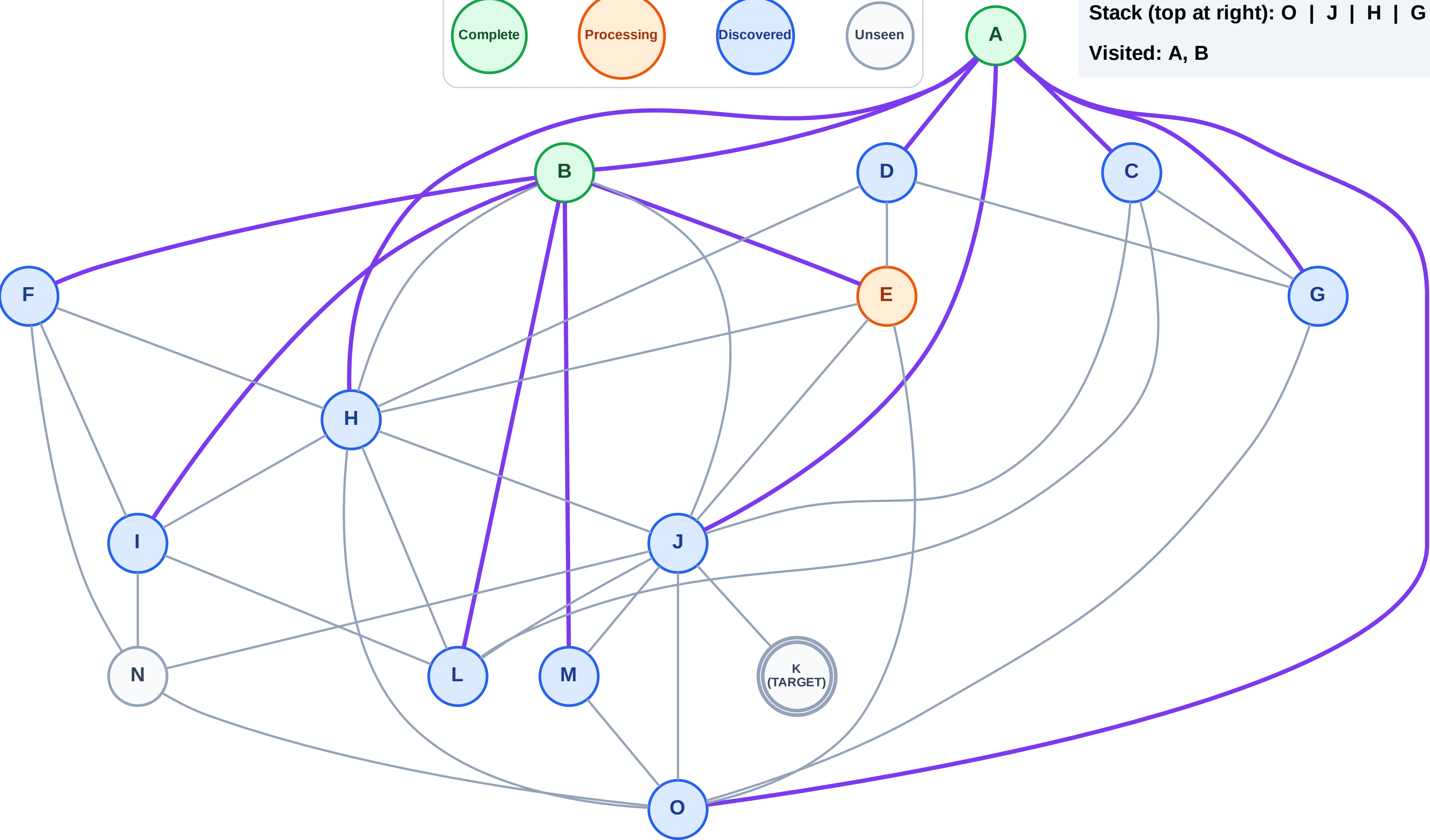
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C | M | L | I | F

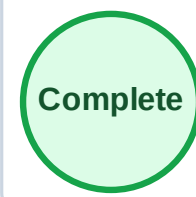
Visited: A, B



# DFS Traversal

Step 7: No new neighbors from E

## Legend



Complete



Processing



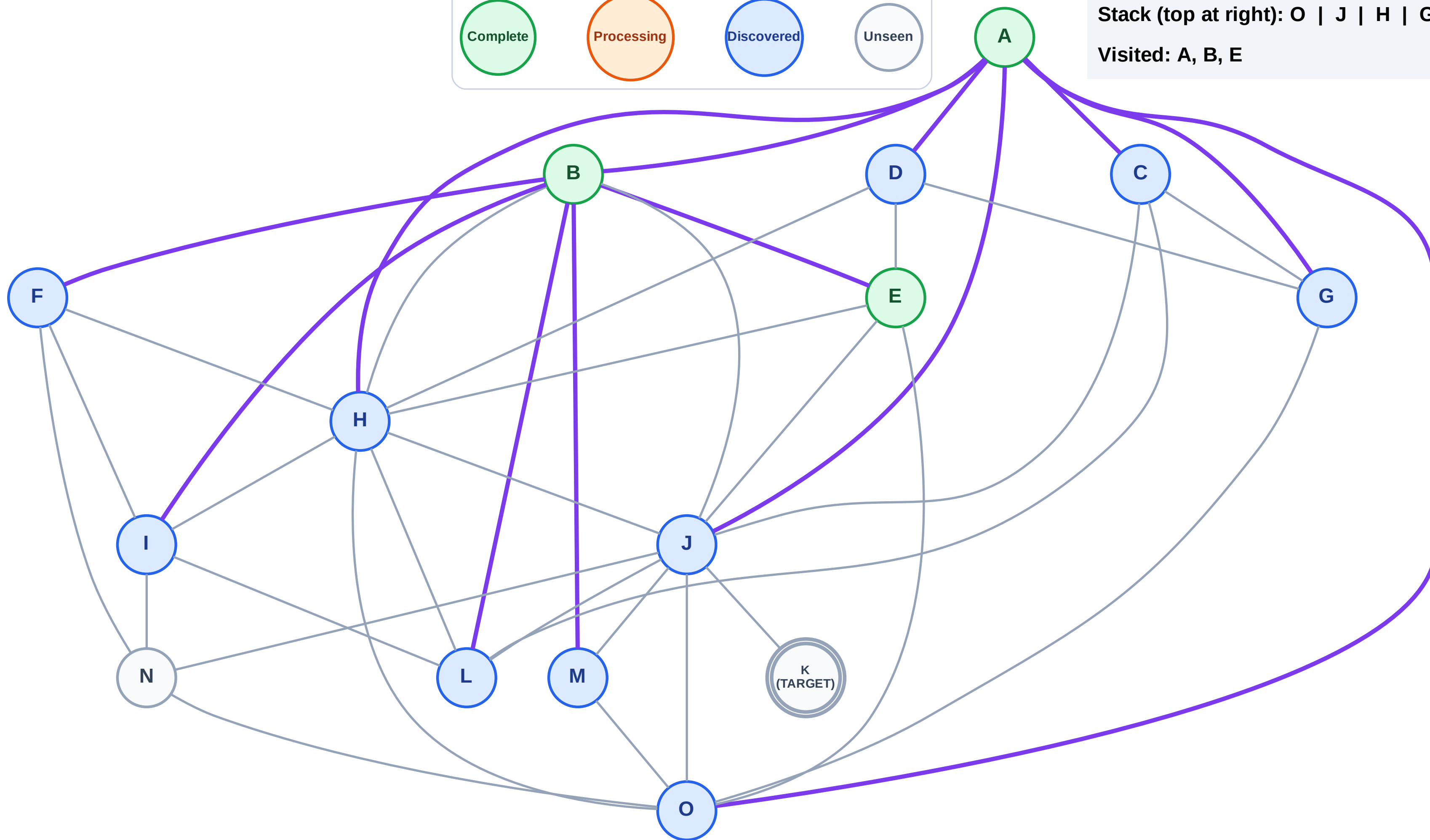
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C | M | L | I | F

Visited: A, B, E



# DFS Traversal

Step 8: Process node F

Legend

Complete

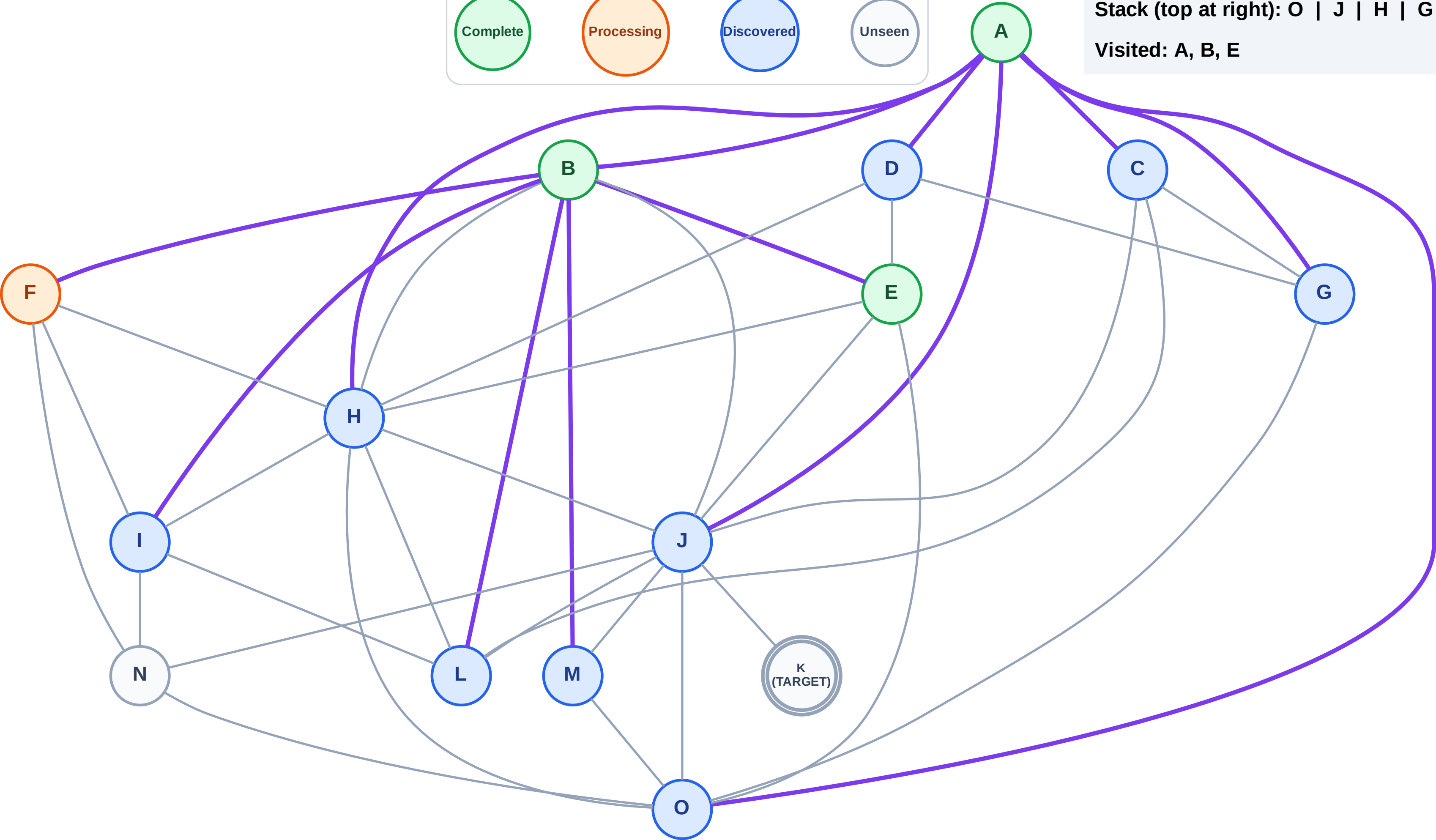
Processing

Discovered

Unseen

Stack (top at right): O | J | H | G | D | C | M | L | I

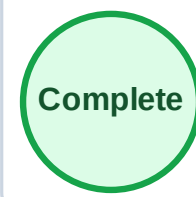
Visited: A, B, E



# DFS Traversal

Step 9: Discover N from F

## Legend



Complete



Processing



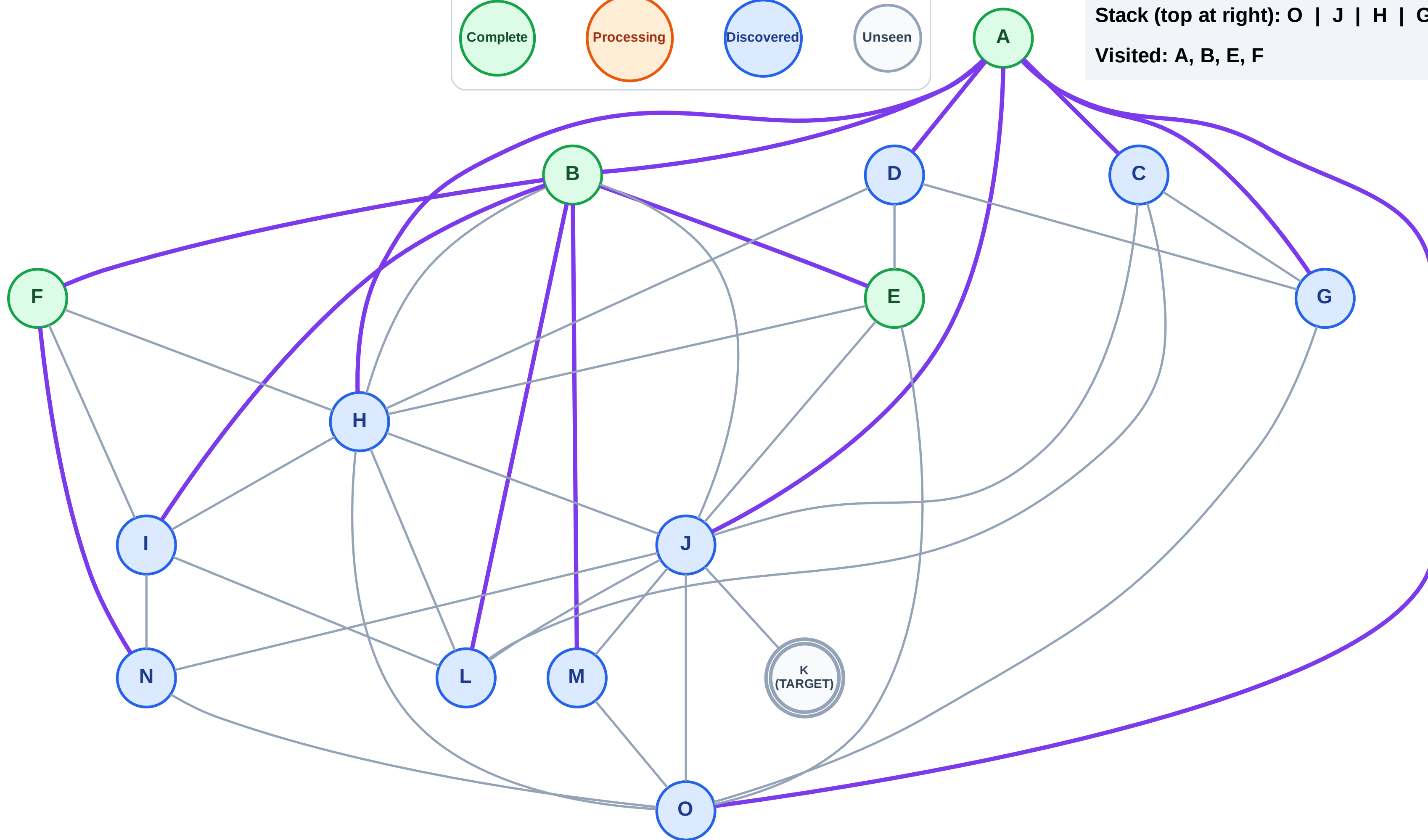
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C | M | L | I | N

Visited: A, B, E, F



# DFS Traversal

Step 10: Process node N

Legend

Complete

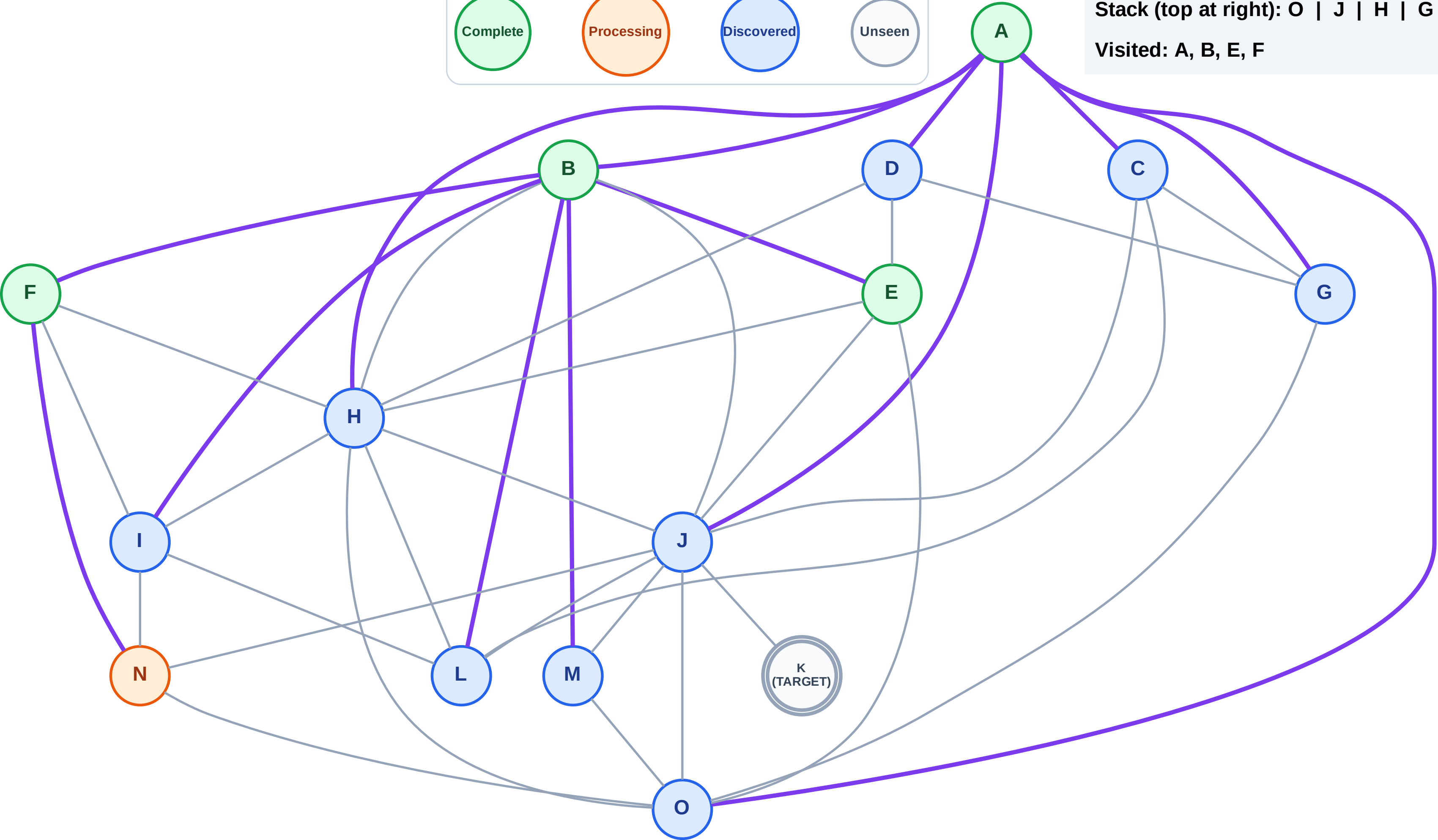
Processing

Discovered

Unseen

Stack (top at right): O | J | H | G | D | C | M | L | I

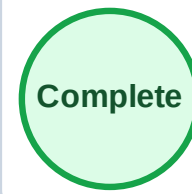
Visited: A, B, E, F



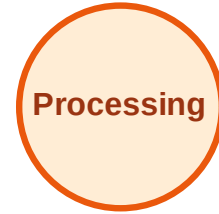
# DFS Traversal

Step 11: No new neighbors from N

## Legend



Complete



Processing



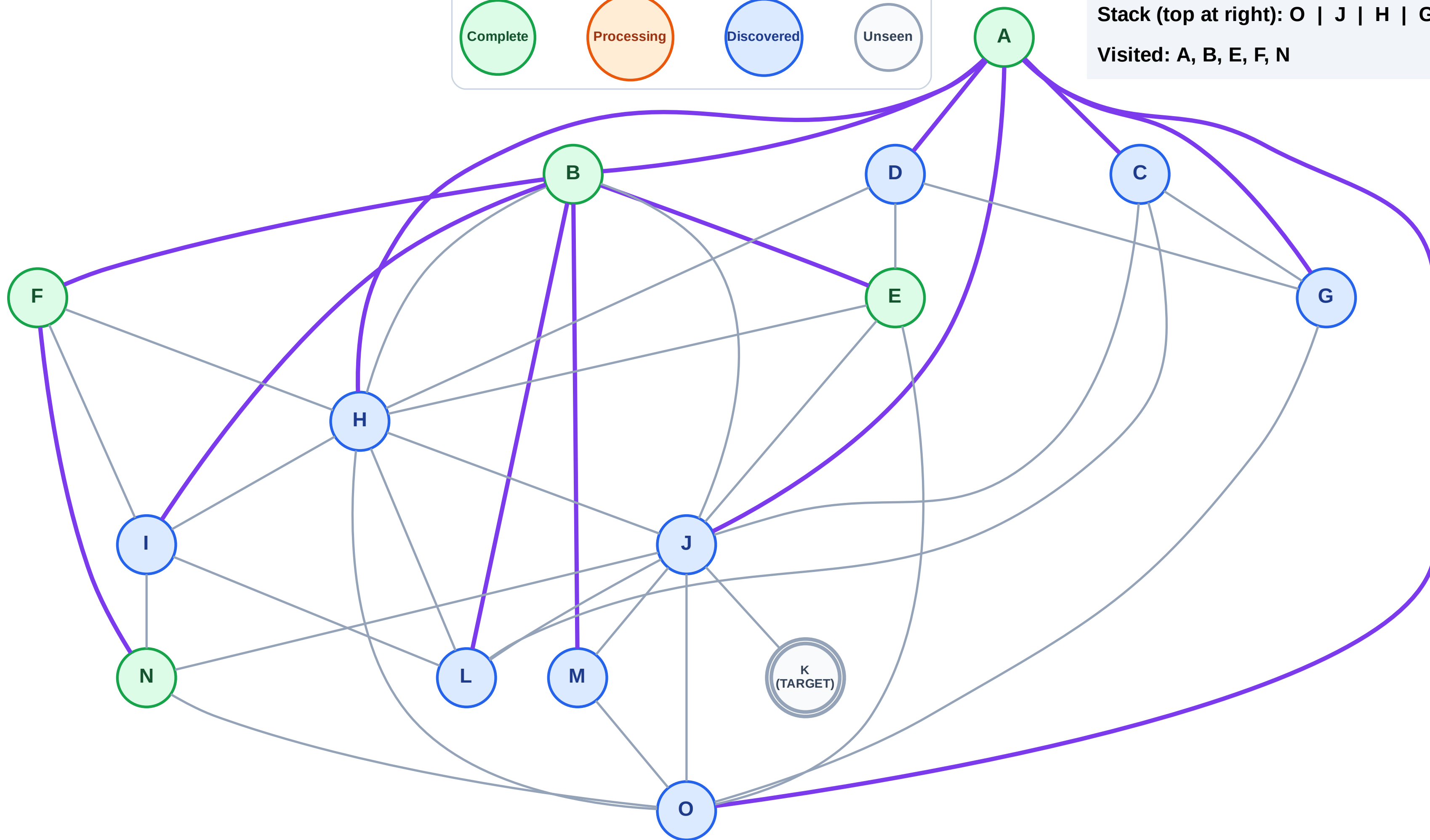
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C | M | L | I

Visited: A, B, E, F, N



# DFS Traversal

Step 12: Process node I

## Legend



Complete



Processing



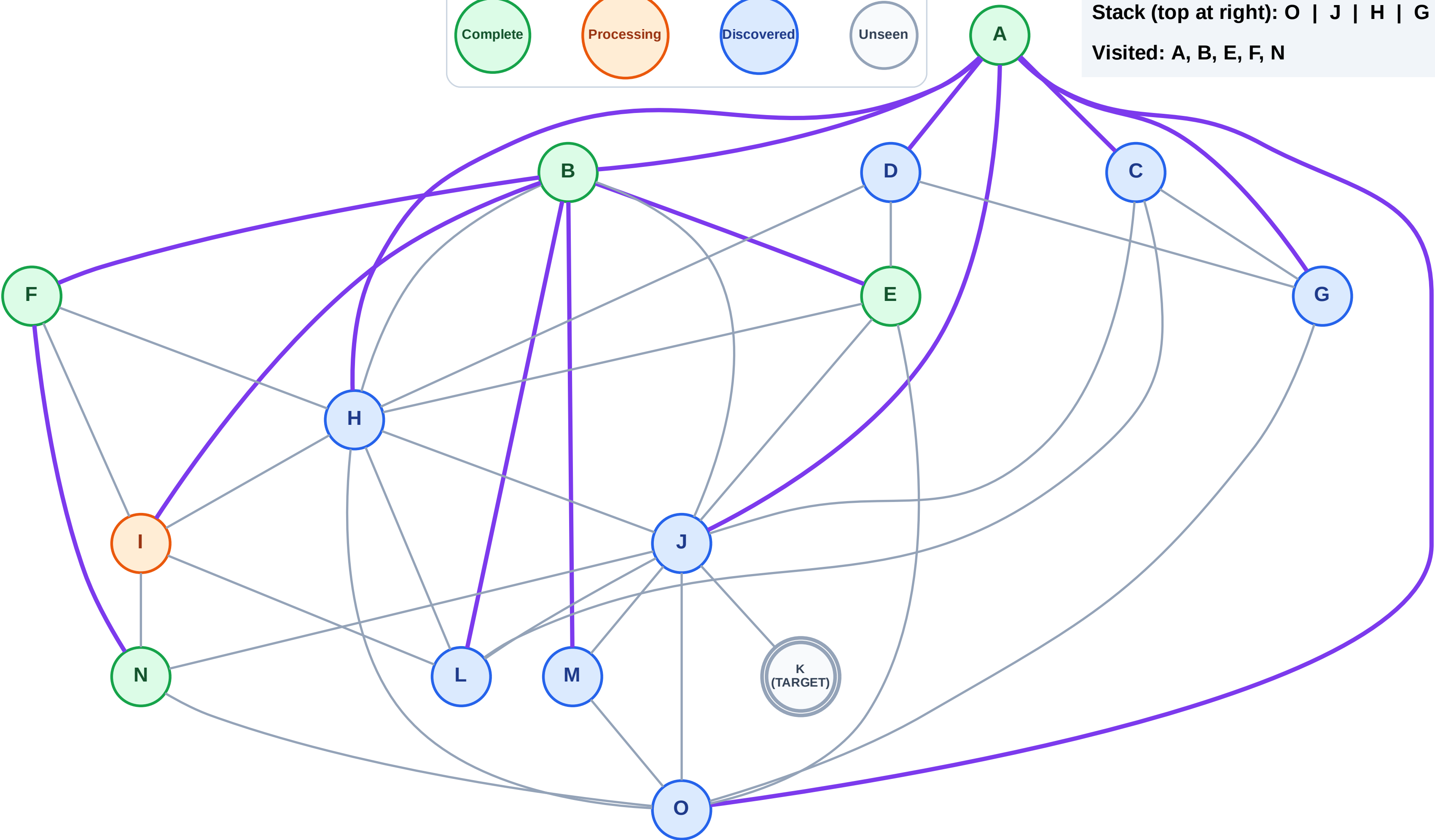
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C | M | L

Visited: A, B, E, F, N



# DFS Traversal

Step 13: No new neighbors from I

## Legend



Complete



Processing



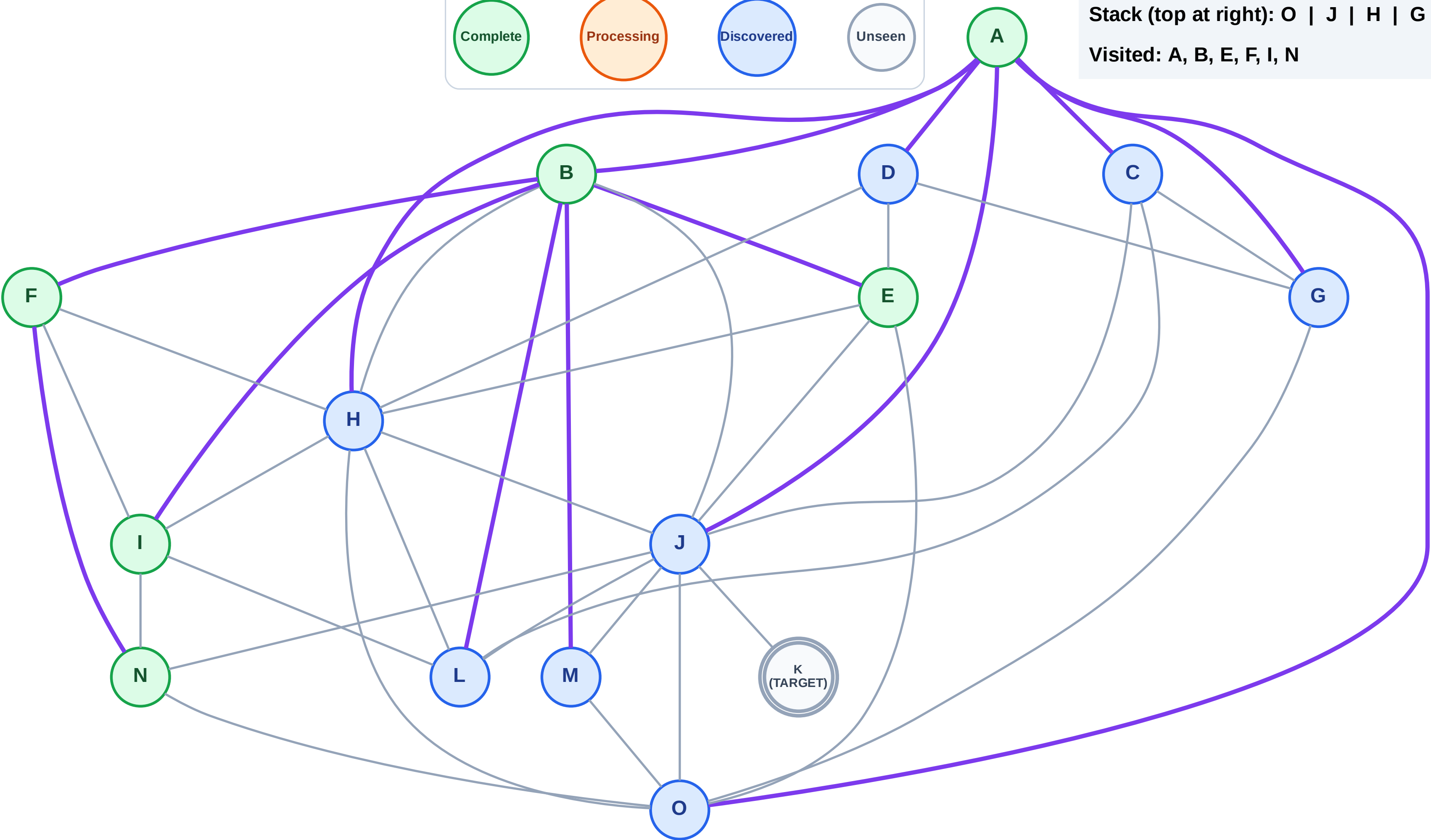
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C | M | L

Visited: A, B, E, F, I, N



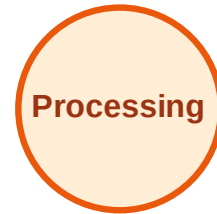
# DFS Traversal

Step 14: Process node L

## Legend



Complete



Processing



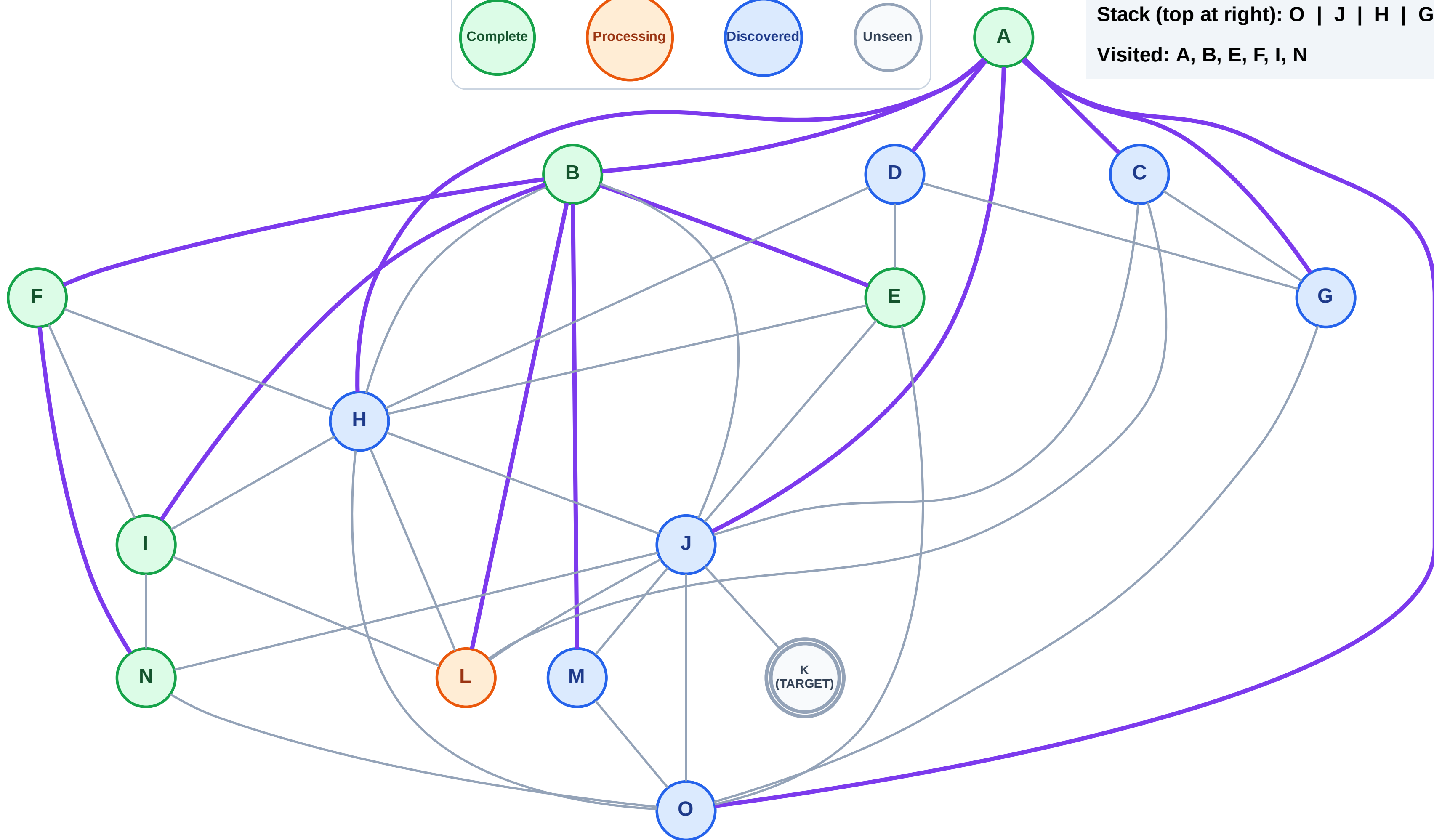
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C | M

Visited: A, B, E, F, I, N



# DFS Traversal

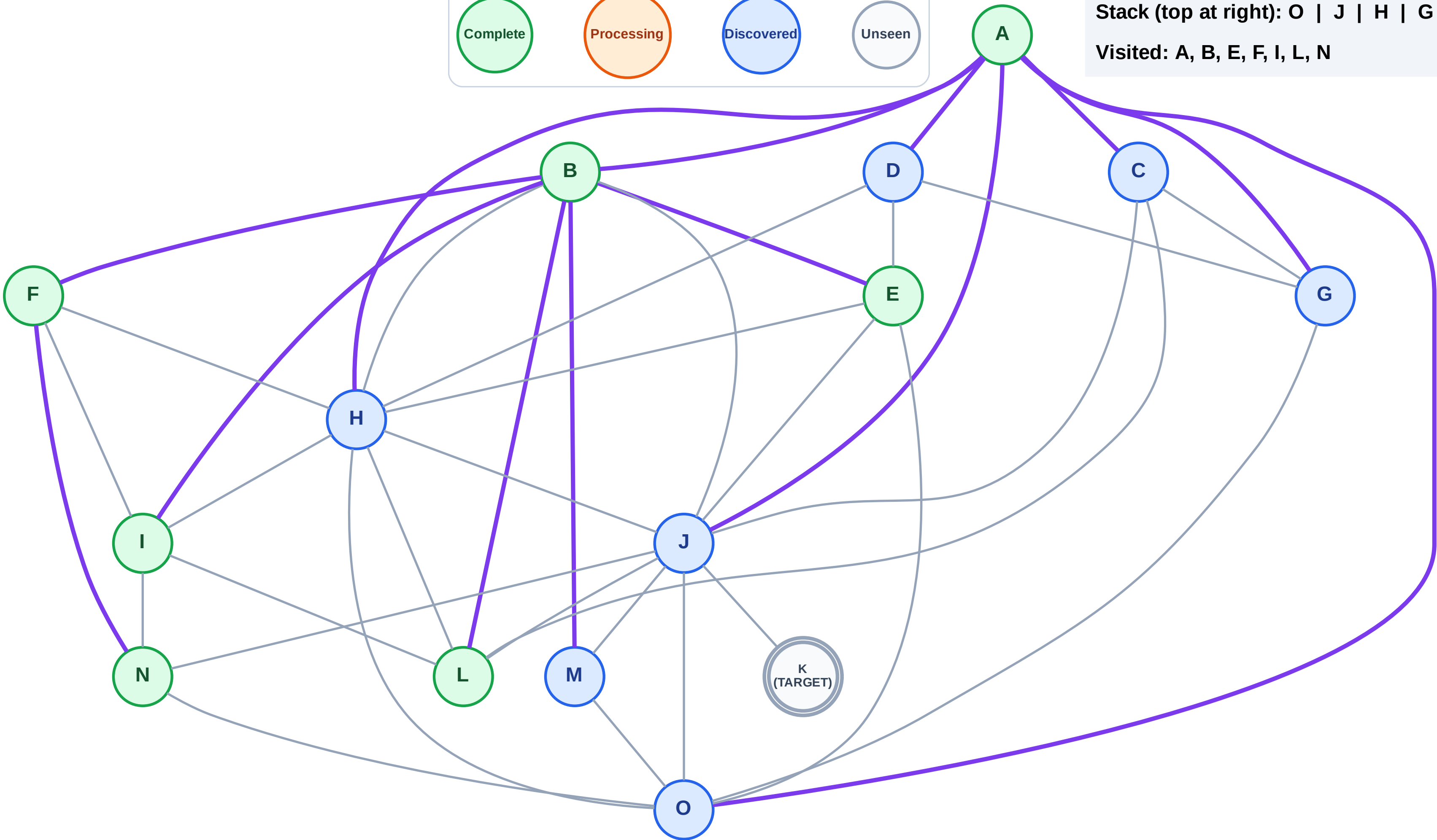
Step 15: No new neighbors from L

## Legend



Stack (top at right): O | J | H | G | D | C | M

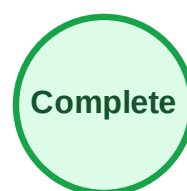
Visited: A, B, E, F, I, L, N



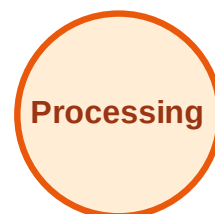
# DFS Traversal

Step 16: Process node M

## Legend



Complete



Processing



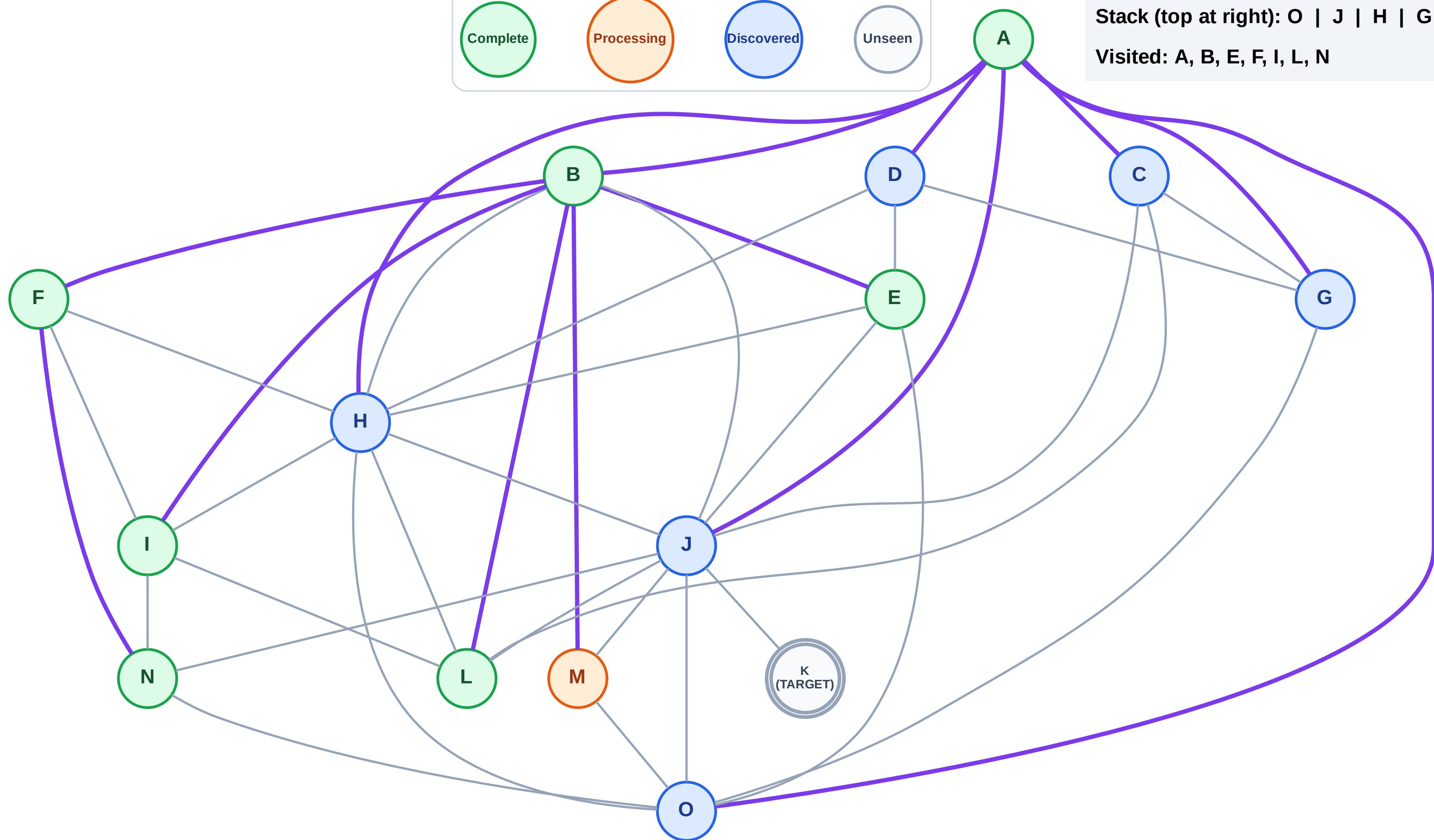
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C

Visited: A, B, E, F, I, L, N



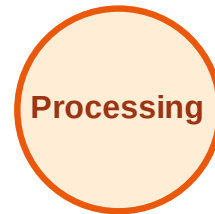
# DFS Traversal

Step 17: No new neighbors from M

## Legend



Complete



Processing



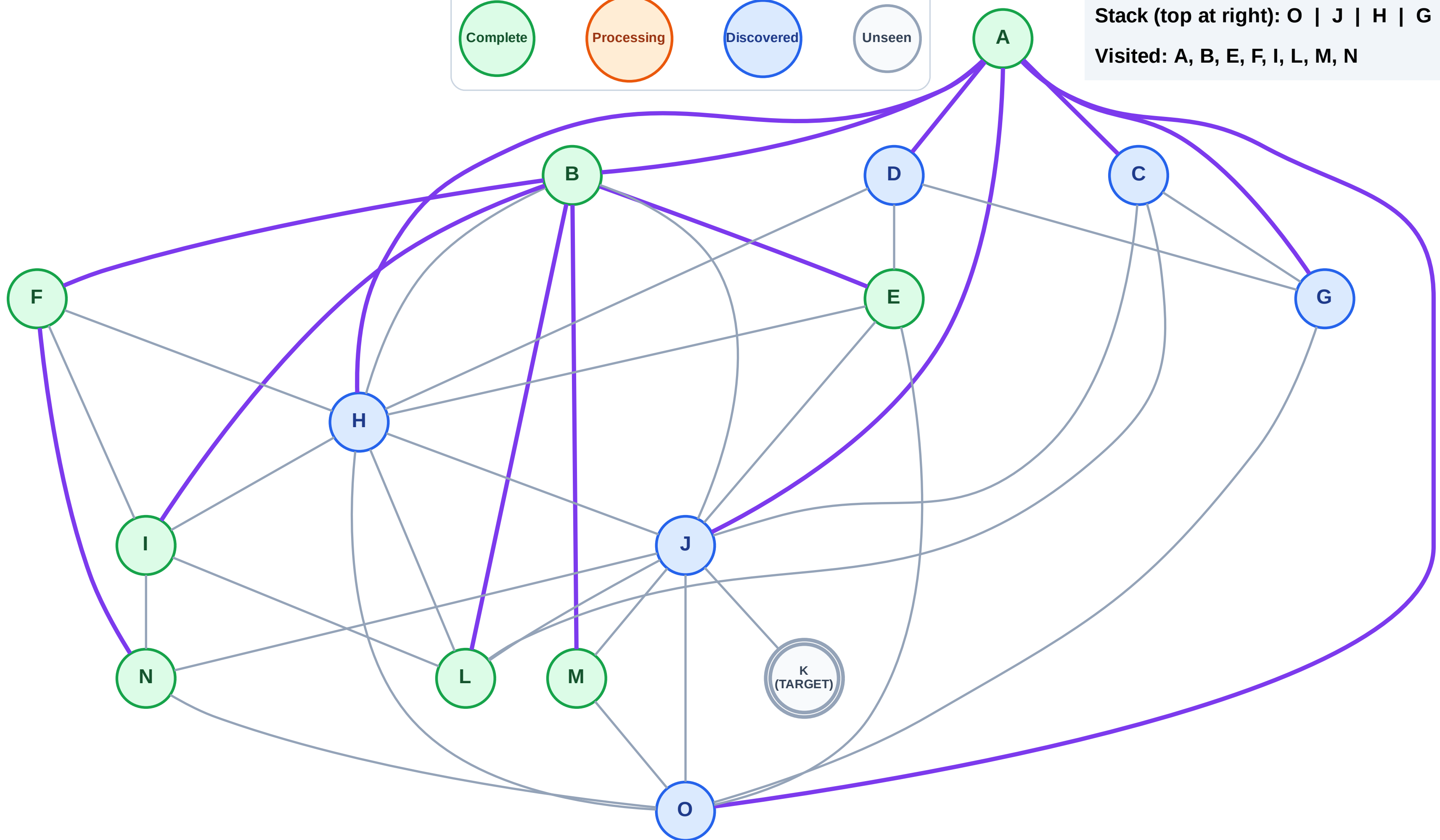
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C

Visited: A, B, E, F, I, L, M, N



# DFS Traversal

Step 18: Process node C

Legend

Complete

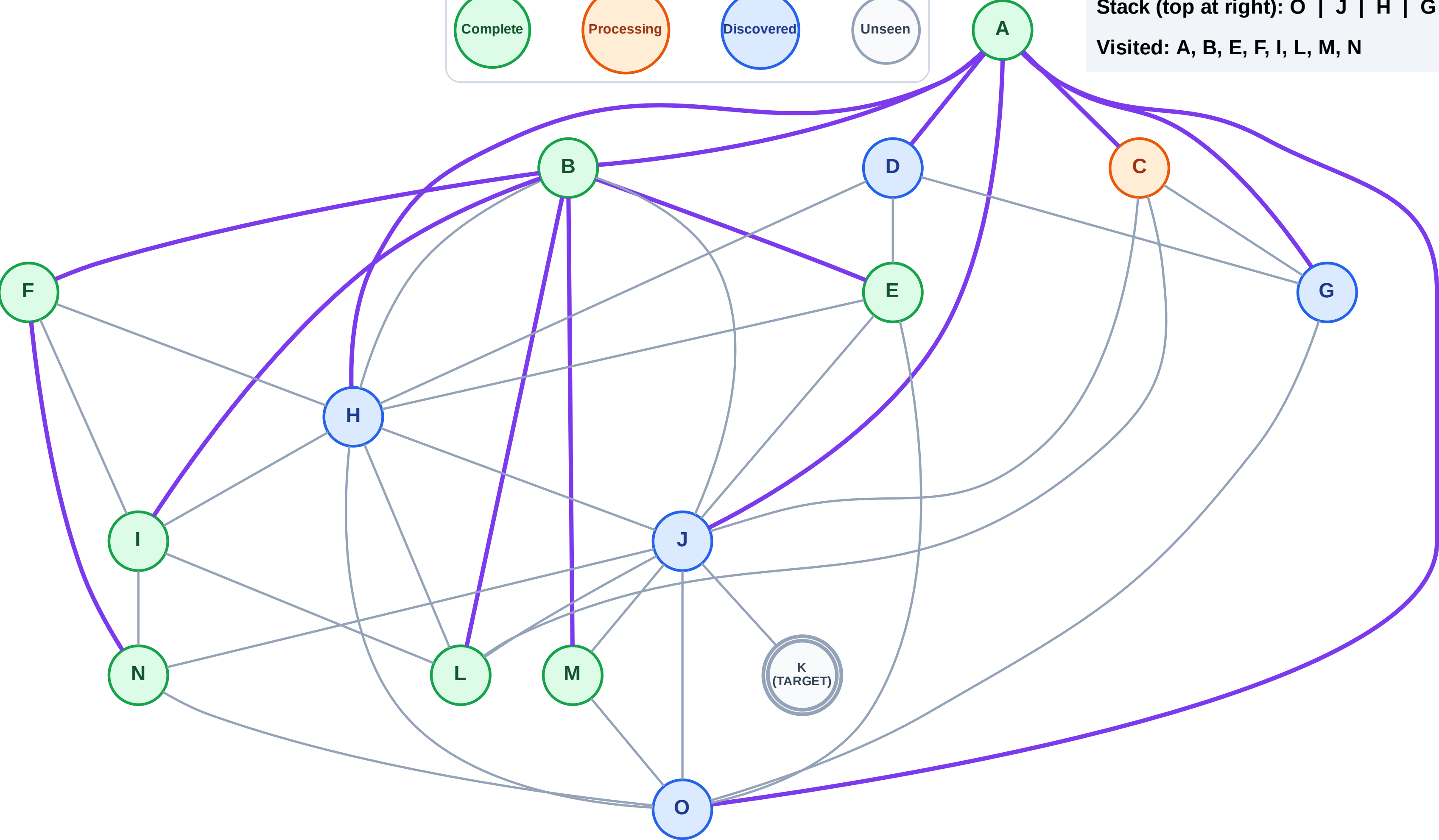
Processing

Discovered

Unseen

Stack (top at right): O | J | H | G | D

Visited: A, B, E, F, I, L, M, N



# DFS Traversal

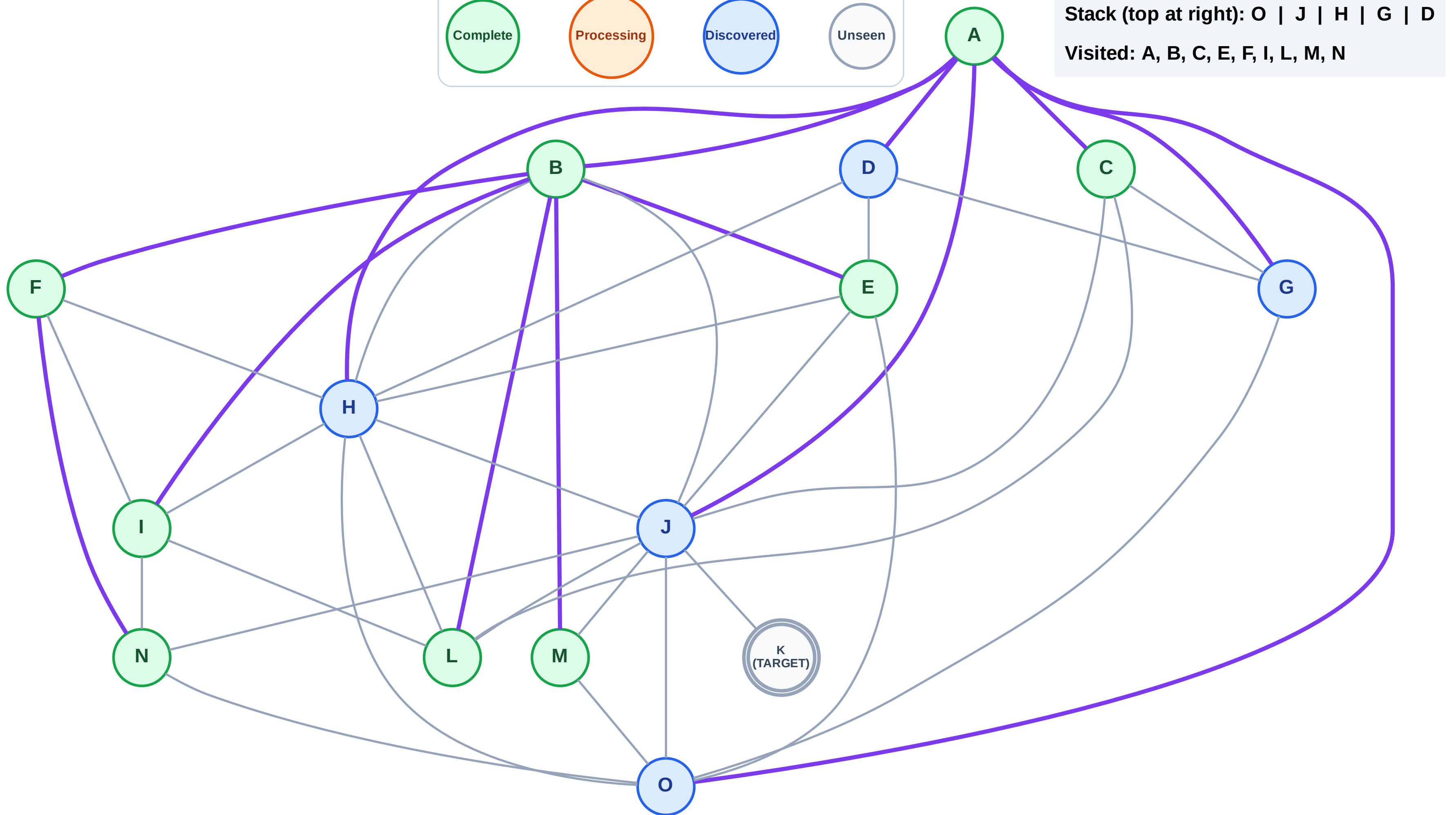
Step 19: No new neighbors from C

## Legend



Stack (top at right): O | J | H | G | D

Visited: A, B, C, E, F, I, L, M, N



# DFS Traversal

Step 20: Process node D

Legend

Complete

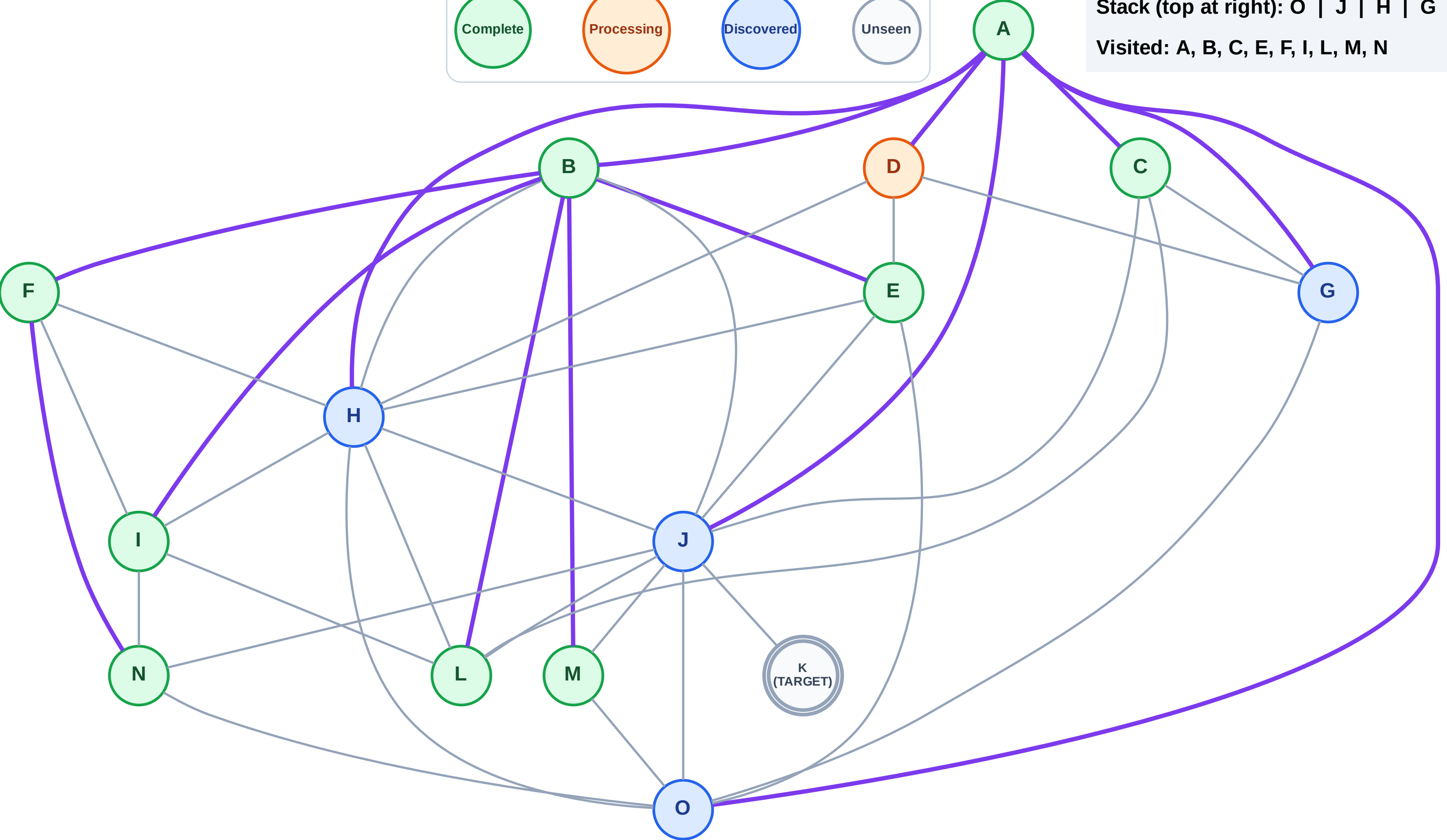
Processing

Discovered

Unseen

Stack (top at right): O | J | H | G

Visited: A, B, C, E, F, I, L, M, N



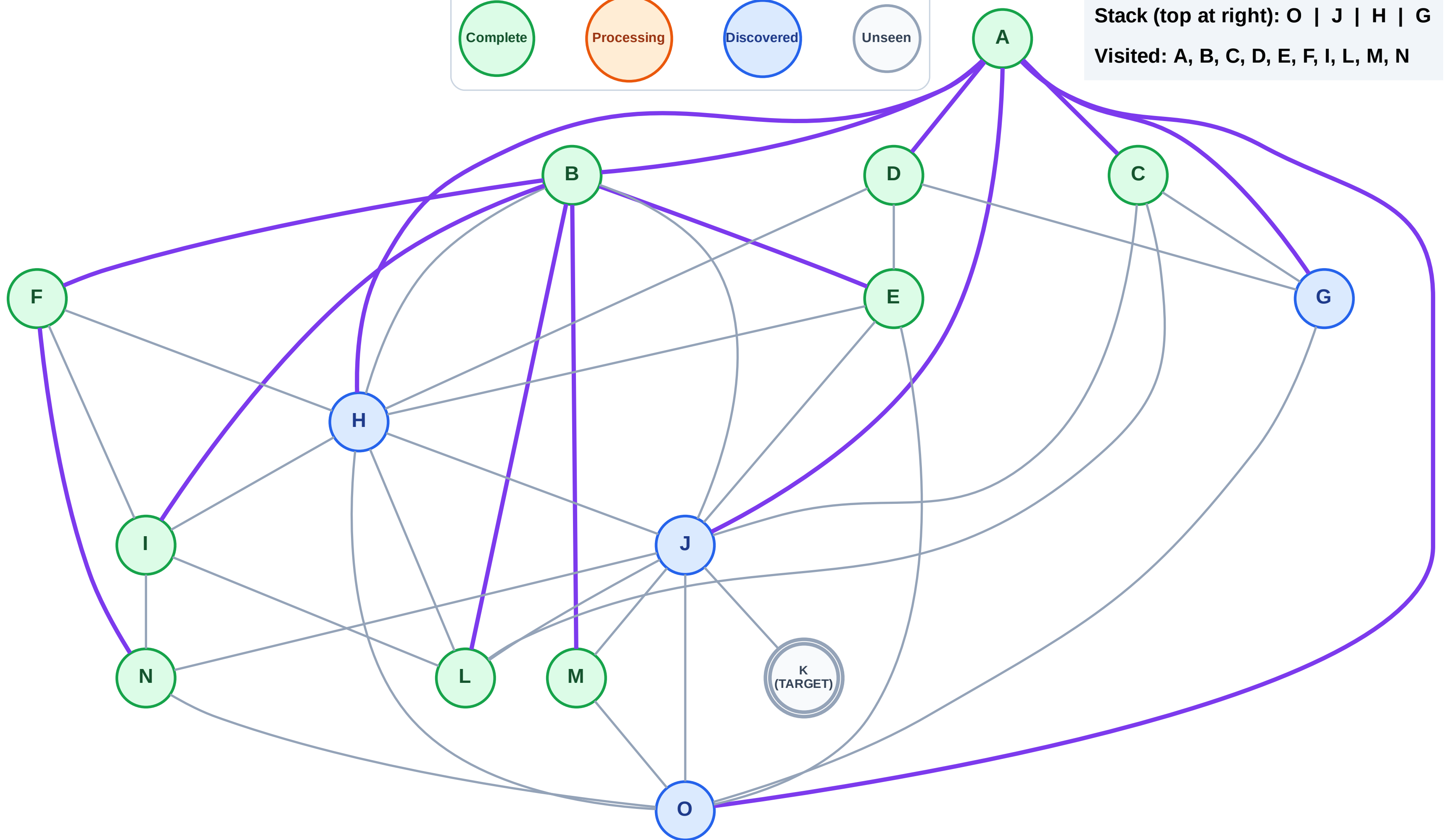
# DFS Traversal

Step 21: No new neighbors from D

## Legend



Stack (top at right): O | J | H | G  
Visited: A, B, C, D, E, F, I, L, M, N



# DFS Traversal

Step 22: Process node G

Legend

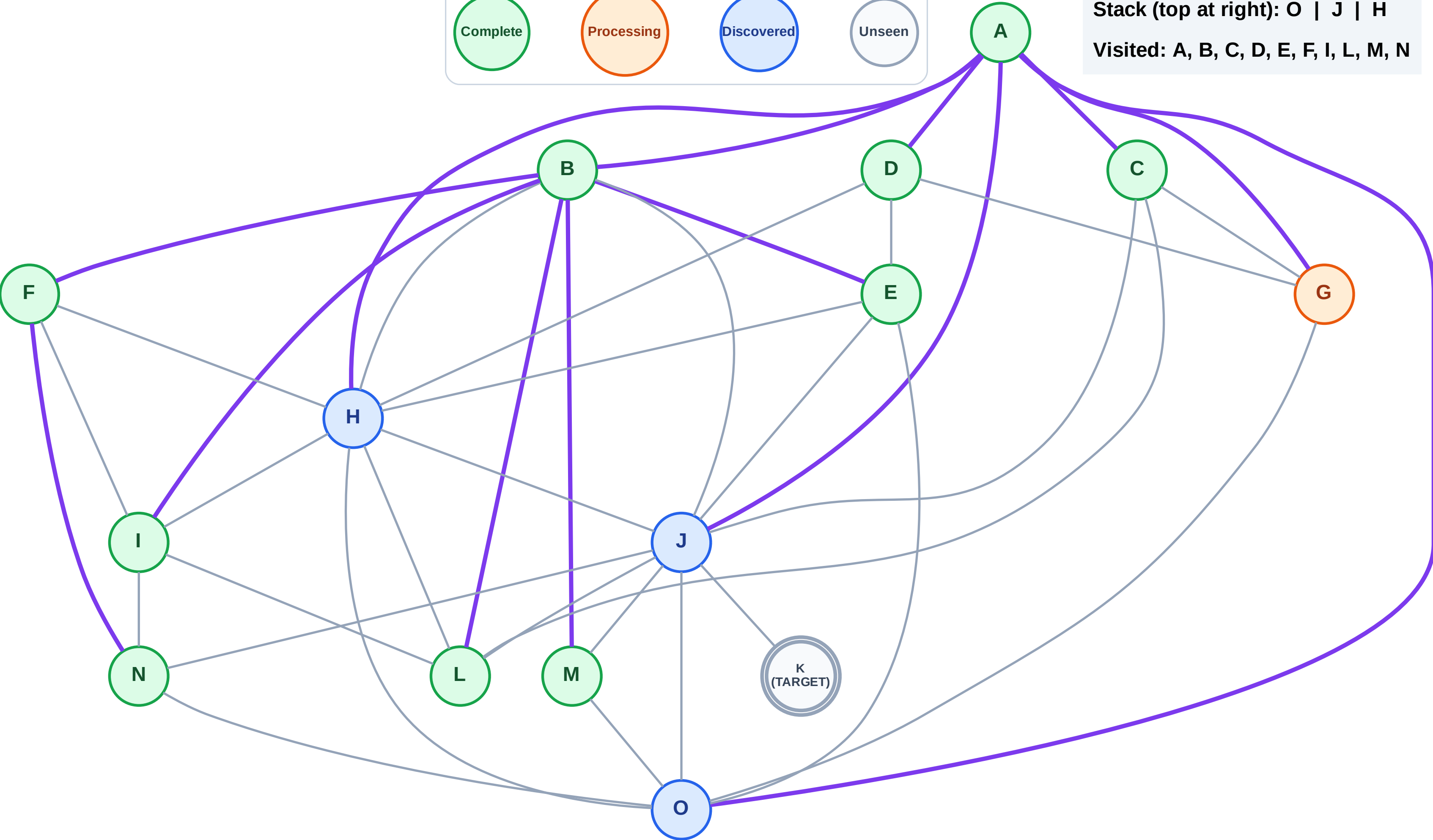
Complete

Processing

Discovered

Unseen

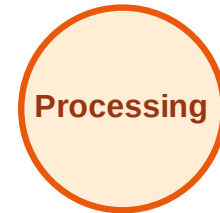
Stack (top at right): O | J | H  
Visited: A, B, C, D, E, F, I, L, M, N



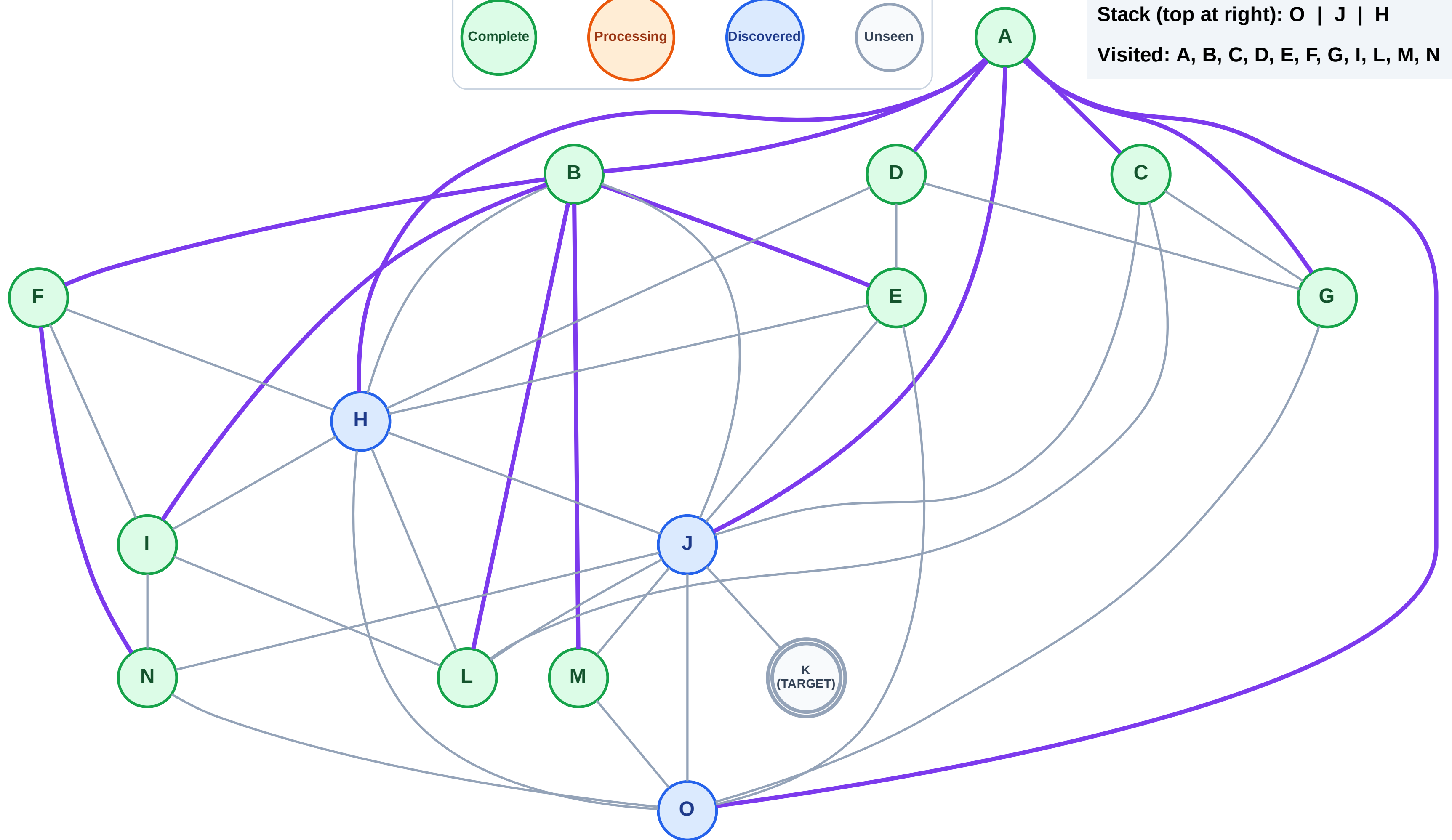
# DFS Traversal

Step 23: No new neighbors from G

## Legend



Stack (top at right): O | J | H  
Visited: A, B, C, D, E, F, G, I, L, M, N



# DFS Traversal

Step 24: Process node H

Legend

Complete

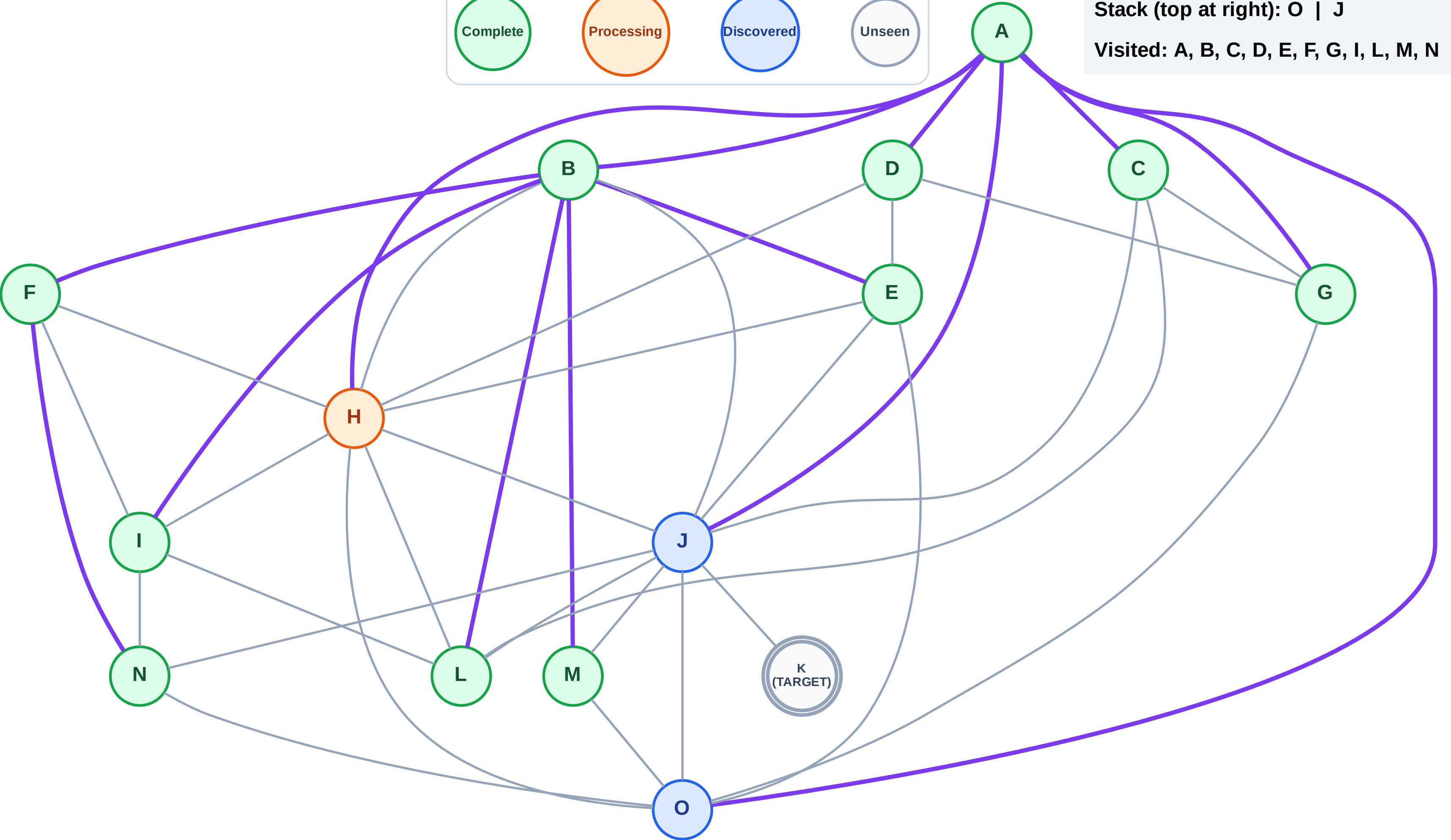
Processing

Discovered

Unseen

Stack (top at right): O | J

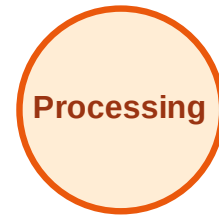
Visited: A, B, C, D, E, F, G, I, L, M, N



# DFS Traversal

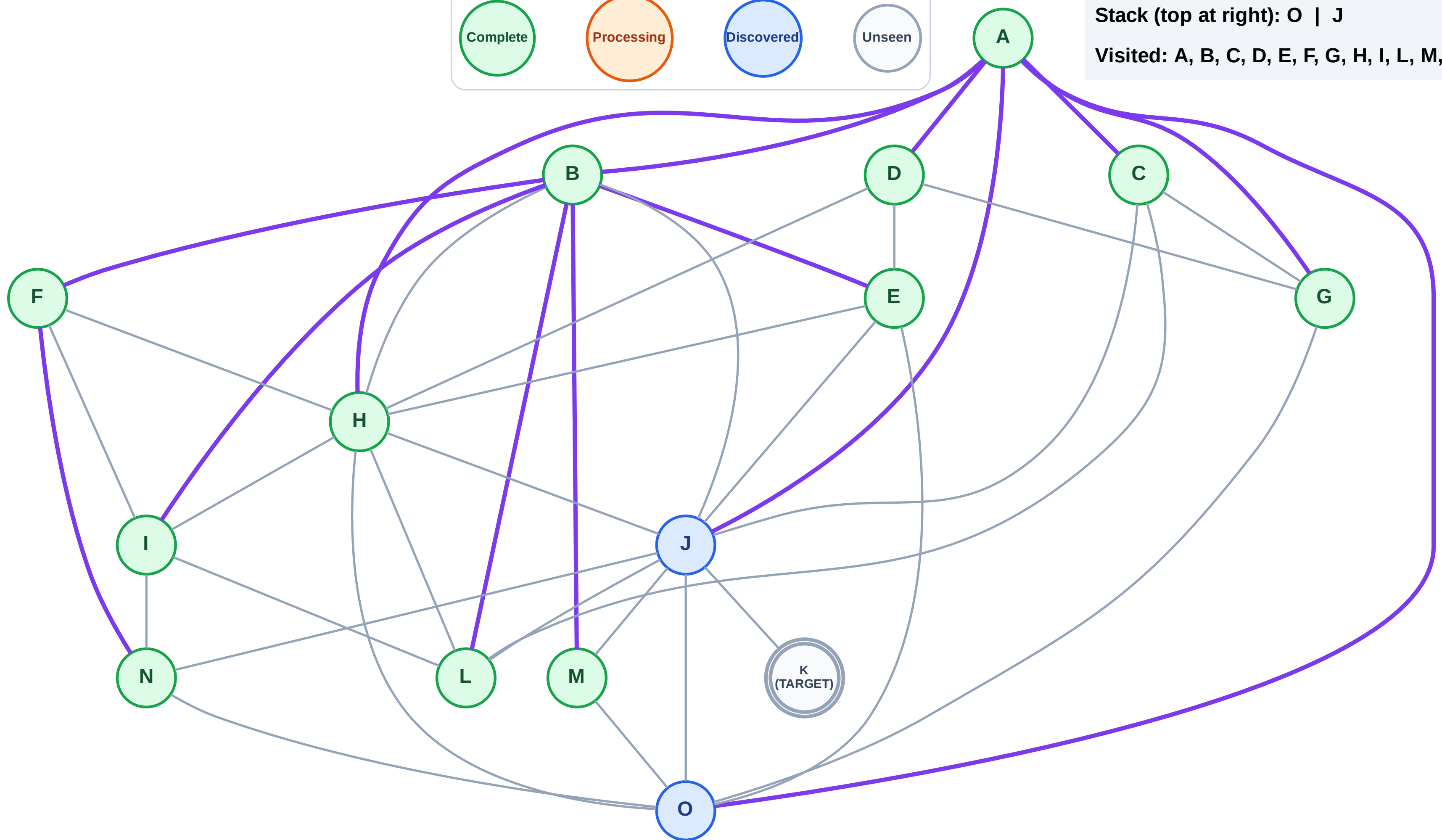
Step 25: No new neighbors from H

## Legend



Stack (top at right): O | J

Visited: A, B, C, D, E, F, G, H, I, L, M, N



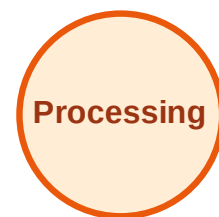
# DFS Traversal

Step 26: Process node J

## Legend



Complete



Processing



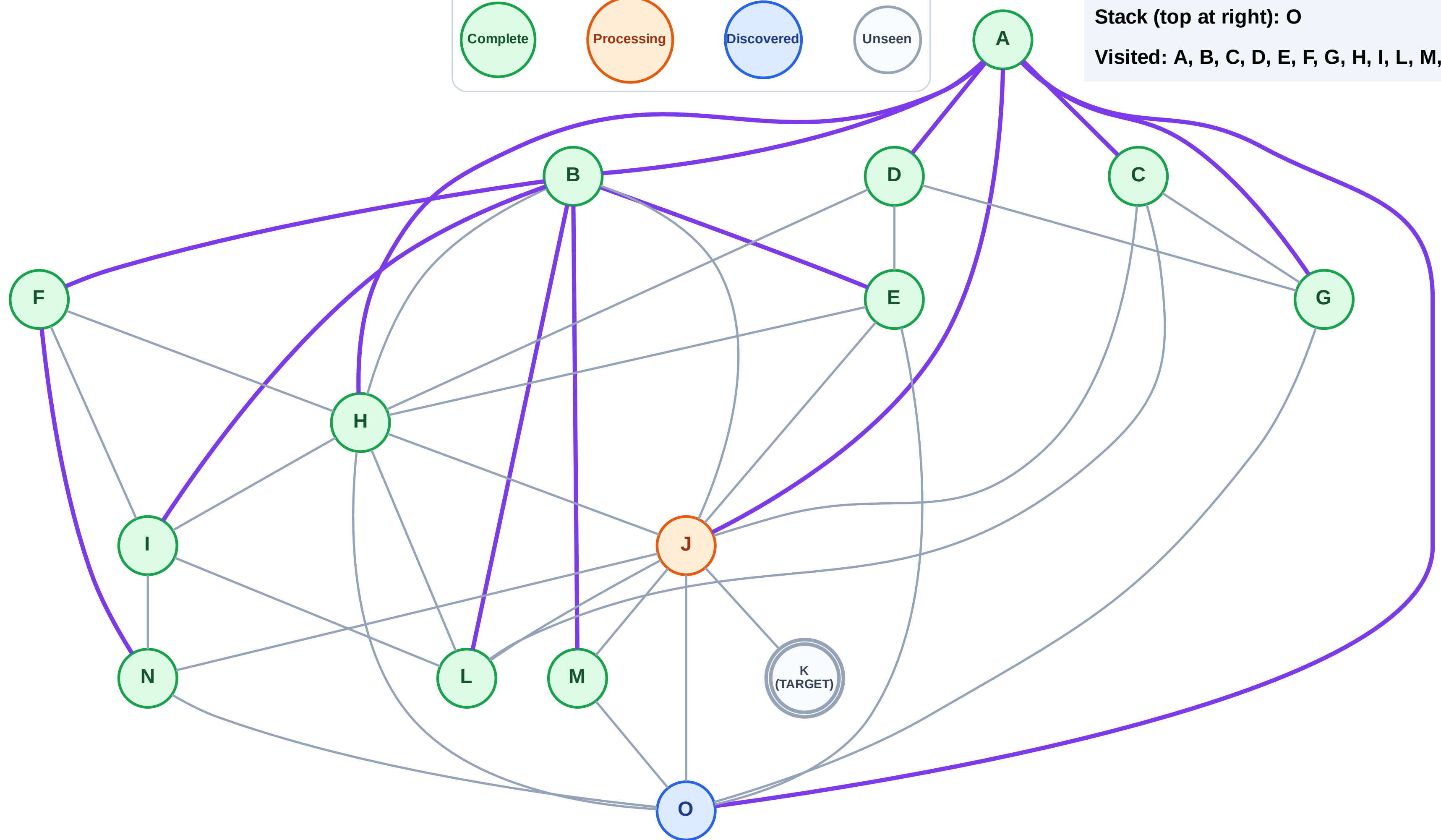
Discovered



Unseen

Stack (top at right): O

Visited: A, B, C, D, E, F, G, H, I, L, M, N



# DFS Traversal

Step 27: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → J → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N

